

Review

Advancing complex product-service systems (CoPSS) design: An integrative literature review

Yiting Zhao^{a,*}, Lihong Zhang^a, Eunice Maytorena-Sanchez^b, Jun Zhang^c

^a Department of Civil Engineering and Management, Faculty of Science and Engineering, The University of Manchester, Oxford Rd, Manchester, M13 9PL, UK

^b Alliance Manchester Business School, The University of Manchester, Oxford Rd, Manchester, M13 9PL, UK

^c School of Design, Hunan University, Lushan Rd, Changsha, 410082, China

HIGHLIGHTS

- Identifies servitisation, digitalisation, sustainability as emerging design transitions.
- Combines bibliometric and thematic analysis in an integrative literature review.
- Synthesises CoPSS design themes across social, technical, ecological dimensions.
- Reveals fragmented integration across triple transitions in CoPSS literature.
- Develops a framework to guide future holistic and adaptive CoPSS design research.

ARTICLE INFO

Keywords:

Complex Product–Service Systems (CoPSS)
Product–Service Systems (PSS)
Social–technical–ecological system design
Integrative Literature Review (ILR)

ABSTRACT

As industries grapple with mounting multidisciplinary design challenges, Complex Product–Service Systems (CoPSS) emerge as a strategic response for delivering value through integrated social, technical, and ecological systems. Existing Product–Service Systems (PSS) frameworks are insufficient to manage these overlapping pressures. While Smart PSS, Sustainable PSS, and Industrial PSS address components of this shift, they do so in isolation. CoPSS has been increasingly described as a system-of-systems design approach that integrates digital, service, and ecological strategies to enhance systemic value.

This paper consolidates CoPSS design literature to clarify the field, map its evolution, and propose an integrative framework for future research. It adopts an integrative literature review methodology, combining bibliometric and thematic analyses. The review draws on 723 peer-reviewed journal articles, from which 62 CoPSS design studies were selected through structured screening and full-text appraisal for in-depth thematic analysis.

Key findings reveal three design clusters that correspond to the emerging transitions in servitisation, digitalisation, and sustainability, with limited integration. Fewer than 1 % of studies addressed all three transitions jointly. CoPSS design themes span technical, social, and ecological domains, including system modelling, co-creation, and circular design. However, integration across these domains remains rare. A conceptual framework maps dominant research themes, methods, theoretical lenses, and proposed directions across intersectional dimensions, highlighting convergence patterns and structural gaps.

This is the first integrative review to synthesise CoPSS design across the triple transitions. It advances theoretical clarity, methodological diversity, and strategic direction, and offers an actionable framework to guide holistic, adaptable, and sustainable system design.

1. Introduction

Product–Service Systems (PSS) were originally developed as business

models and design strategies that combine tangible products with intangible services to deliver enhanced value and user satisfaction (Tukker, 2004). By shifting from ownership to access, and from isolated

* Corresponding author.

E-mail addresses: yiting.zhao@manchester.ac.uk (Y. Zhao), lihong.zhang@manchester.ac.uk (L. Zhang), EuniceMaytorena-sanchez@manchester.ac.uk (E. Maytorena-Sanchez), zhangjun@hnu.edu.cn (J. Zhang).

<https://doi.org/10.1016/j.jclepro.2025.146690>

Received 17 May 2024; Received in revised form 1 August 2025; Accepted 19 September 2025

Available online 23 September 2025

0959-6526/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

artefacts to integrated offerings, PSS challenged conventional product-centric design by embedding service functionality within the use context. Over the past two decades, the concept has been widely applied across sectors including manufacturing, mobility, and health-care, as a means of delivering more tailored, efficient, and responsive solutions (Mourtzis et al., 2018a; Rahmati et al., 2021; Roehrich and Lewis, 2010).

As PSS applications have expanded, so too have the challenges they pose for system design. The increasing need to coordinate across distributed actors, adaptive technologies, and system-wide performance goals (Hallstedt et al., 2020; Isaksson et al., 2018) has led to the emergence of Complex Product–Service Systems (CoPSS). CoPSS refer to integrated configurations of products, services, and systems that leverage advanced technologies and business networks to maximise value across social, technical, and ecological domains (Boess, 2022; Roman et al., 2023; Tsunetomo et al., 2022; Yin et al., 2020). They are characterised by modular architectures, multi-actor coordination, real-time data flows, and long-term responsiveness to both operational and societal objectives (Sassanelli et al., 2018). Rather than merely expanding the scope of PSS, CoPSS represent a qualitative shift in design thinking. They require ongoing collaboration, value alignment, and system-wide governance across actors, technologies, and environmental goals (Dehn et al., 2018; Dober et al., 2023; Ringen et al., 2022).

A representative example is the design of a smart building system, which must synchronise energy-efficient hardware, intelligent control platforms, occupant satisfaction, and environmental goals. It involves diverse stakeholders with different expectations, such as engineers, facility managers, users, and platform providers, and demands lifecycle integration from early-stage design to post-occupancy performance (Tsunetomo et al., 2022; Zhao et al., 2021). These requirements exceed the assumptions of early PSS models, pointing to the need for more integrated design strategies capable of managing complexity across interconnected system layers.

Despite growing interest, the field of CoPSS design remains fragmented. Researchers working within Smart PSS, Sustainable PSS, and Industrial PSS streams have each contributed distinct perspectives, but often operate in disciplinary silos, with limited interaction or synthesis. Smart PSS emphasises digital connectivity and data-driven services but often lacks mechanisms for addressing broader system-level impacts (Dong et al., 2023). Sustainable PSS promotes environmental considerations but typically isolates sustainability as a downstream evaluation rather than an integrated design concern (Vezzoli et al., 2015). Industrial PSS has advanced the operational and organisational dimensions of service integration, but with limited attention to ecological trade-offs or participatory processes (Meier et al., 2010). These streams address aspects of the design challenge, but few studies engage with the cross-domain complexity that defines CoPSS in practice.

This fragmentation has created several unresolved gaps in CoPSS design research. The literature offers limited insight into how dispersed studies are beginning to align around shared challenges, or how this convergence shapes an emerging trajectory. There is no systematic synthesis of research themes, methods, or theoretical perspectives, nor a structured foundation to support integration across social, technical, and ecological dimensions. As a result, contributions remain scattered, limiting the field's capacity to build cumulative insight or offer coherent guidance.

This paper addresses these issues by providing the first integrative review of CoPSS design research. It adopts an evidence-led strategy to examine how CoPSS has emerged in the literature, how its design has been investigated, and what future research is needed to support its development. The review employs an Integrative Literature Review (ILR) methodology (Torraco, 2016), which is well-suited for emerging and interdisciplinary fields of research. ILR enables the combination of bibliometric analysis, which is used to track thematic patterns and field development, with qualitative synthesis, which is used to interpret meaning, methods, and theoretical framing across diverse studies. This

approach supports both structural mapping and interpretive depth, enabling the derivation of new conceptual frameworks from diverse and fragmented literatures (Yin et al., 2020).

This review aims to consolidate conceptual development in CoPSS design and to examine how the field has been studied to date. To achieve this, the review pursues three specific objectives: (i) to examine how CoPSS design research has evolved within the broader PSS design literature; (ii) to synthesise the current state of CoPSS design research, including key research themes, theoretical lenses, methodological approaches, and literature-proposed directions; and (iii) to develop an integrative framework that maps convergence patterns and structural gaps across social, technical, and ecological dimensions to inform future inquiry.

These objectives are addressed through three guiding research questions (RQs):

- RQ1: How has CoPSS design research evolved within the broader PSS literature, and what dominant thematic transitions have emerged?
- RQ2: What is the current landscape of CoPSS design research in terms of its dominant research themes, employed methodological approaches, mobilised theoretical foundations, and research directions proposed by the literature?
- RQ3: How can an integrative framework synthesise these findings and guide future research on CoPSS design?

The review is conducted in three stages. In the first stage, a bibliometric analysis of 723 peer-reviewed PSS design articles is used to trace the field's development toward CoPSS and to identify structural clusters (RQ1). The patterns and clusters identified in this stage inform both the article selection and the thematic interpretation in the second stage, which involves a structured screening and full-text analysis of 62 CoPSS design studies (RQ2) to synthesise the landscape of CoPSS design research. The third stage constructs an integrative framework based on these findings, mapping convergence patterns and gaps across social, technical, and ecological dimensions to inform future research (RQ3). This staged approach ensures transparency and coherence in the review process, addressing common concerns about replicability and interpretive bias in literature reviews (Yin et al., 2020).

This review makes three key contributions. First, it identifies a structural convergence of servitisation, digitalisation, and sustainability within the PSS design literature, as revealed by the bibliometric analysis, signalling a shift towards more integrated and systemic design inquiry. Second, it provides a comprehensive synthesis of the current CoPSS design research landscape, identifying dominant research themes, methodological approaches, theoretical framings, and proposed directions. Third, it proposes a novel integrative framework that structures this landscape across social, technical, and ecological dimensions, highlighting both areas of convergence and critical research gaps. By doing so, the review enhances conceptual clarity, methodological structure, and strategic guidance for scholars and practitioners seeking to develop holistic, adaptive, and sustainable CoPSS solutions.

The remainder of this paper is structured as follows. Section 2 outlines the review methodology. Section 3 examines the structural trends and topic clusters in PSS design research using bibliometric analysis. Section 4 synthesises CoPSS design studies through thematic analysis. Section 5 presents an integrative framework. Section 6 discusses the contributions, implications, and limitations, and concludes the review.

2. Methodology

This review employed an Integrative Literature Review (ILR) methodology, a form of knowledge synthesis that critically reviews and integrates representative literature, allowing new conceptual frameworks and insights to emerge (Torraco, 2016). ILR is particularly suited to emerging, interdisciplinary fields like CoPSS design, as it enables a holistic understanding by integrating diverse perspectives through the use

of broad sampling strategies and the combination of various literature analysis methods. This approach aligns well with the research questions on trends (RQ1) and the current state (RQ2) of CoPSS design research (Toronto and Remington, 2020).

This study aims to examine the evolution and current state of CoPSS design research and formulate an agenda for future research. This aim requires both a comprehensive view of the publication landscape and a fine-grained interpretative lens. ILR is therefore the most appropriate choice, as it allows quantitative mapping to trace evolution while simultaneously supporting qualitative interpretation to build new conceptual frameworks. A systematic review would impose inclusion rules that are too restrictive for an emergent, cross-disciplinary field (Rashid et al., 2019), whereas a purely bibliometric review would chart patterns but leave the underlying meanings unexplored (Zupic and Čater, 2015). ILR strikes the balance required by our dual objective: to track the trajectory of CoPSS design research and to explain what that trajectory means for future inquiry.

To address potential methodological limitations inherent in ILR, including the risk of reduced rigour and possible bias due to its broad scope, this study adopts a systematic review process comprising literature search, screening and appraisal, analysis, and results presentation and synthesis. This structured process ensures transparency and reproducibility (Rodríguez et al., 2020; Whittemore and Knafl, 2005; Toronto and Remington, 2020), as illustrated in Fig. 1.

2.1. Literature search

Given the nascent state of CoPSS design literature, this review draws insights from the more established Product–Service Systems (PSS) design literature for essential contextual framing. Two distinct literature pools were constructed: the broader "PSS design" literature and the more focused "CoPSS design" literature (Alcayaga et al., 2019). These pools facilitate comparisons between general trends in PSS design research and specific developments in CoPSS design. Searches were conducted in Scopus, widely acknowledged for its comprehensive coverage of engineering management research, particularly in the PSS domain (Li et al., 2020). The PSS design pool was searched using general keywords that reflect core PSS design components such as product, service, system, and design. The CoPSS design pool was then searched using a more focused search string, informed by the bibliometric analysis of the PSS design pool, which identified the triple transitions of servitisation, digitalisation, and sustainability. Detailed search strings and systematic search outcomes are summarised in Fig. 1.

2.2. Screening and appraisal

The screening and appraisal process was executed in three progressive steps, summarised in Fig. 1. Initially, articles underwent preliminary screening (Screening 1) based on language, source type, and timeframe. Selection criteria included publications from 2003 to 2025, limited to English-language, peer-reviewed journal articles at the final

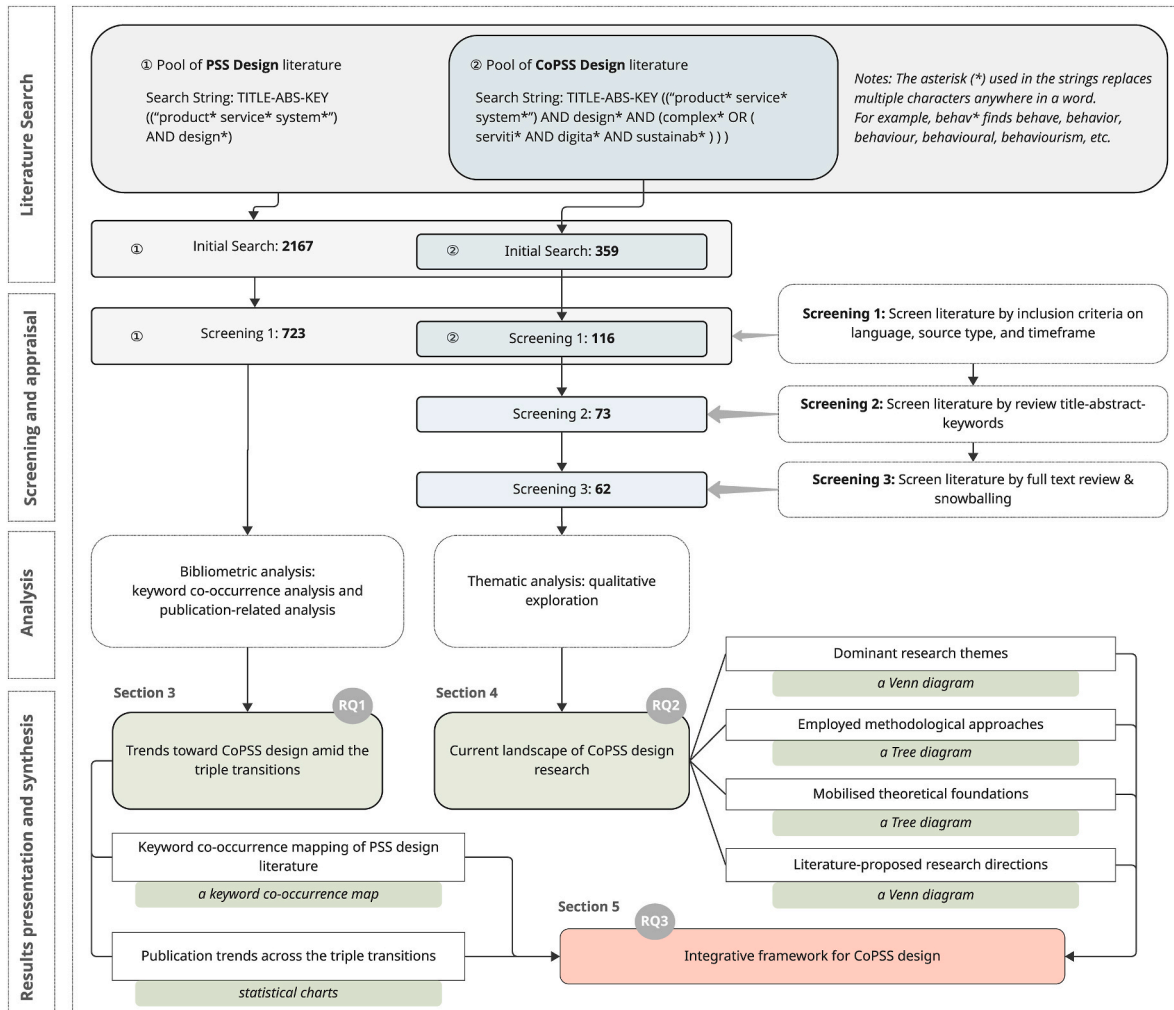


Fig. 1. The integrative literature review process.

publication stage. The year 2003 marks the formal emergence of the PSS concept (Manzini and Vezzoli, 2003). Screening 1 resulted in 723 articles for the broader PSS design literature (for bibliometric analysis), and 116 articles for the more specific CoPSS design literature.

The second screening step (Screening 2) involved reviewing the titles, abstracts, and keywords to confirm explicit relevance to CoPSS design concerning digitalisation, sustainability, servitisation, and complexity. Articles retained at this stage demonstrated clear relevance to CoPSS design and provided methodological or empirical contributions to design practices.

The final screening (Screening 3) involved a full-text review of shortlisted articles to ensure direct relevance to CoPSS design. Each article was critically appraised based on clarity of research design, methodological rigour, explicit focus on CoPSS design processes, tools, methods, or frameworks, and alignment with key transition domains. This stage also included backward and forward snowballing citation tracking to identify additional relevant studies. Ultimately, this resulted in 62 articles forming the core dataset for detailed thematic analysis (last search check conducted on July 09, 2025).

2.3. Analysis

The analysis combined bibliometric analysis and thematic methods to leverage their complementary strengths (Donthu et al., 2021; Rashid et al., 2019). Bibliometric analysis provided quantitative insights into the growth and evolution of CoPSS design within the broader PSS design literature, while thematic analysis offered qualitative exploration into CoPSS design research and elucidated its state-of-the-art developments.

To address RQ1, which aims to uncover the emerging trend of CoPSS design in the PSS design literature, a bibliometric analysis was employed, offering robust quantitative methods suitable for handling large volumes of data (Zupic and Cater, 2015). Two bibliometric techniques were employed. Keyword co-occurrence analysis was specifically selected due to its capability to illustrate interconnected concepts within the literature, identifying clusters of frequently co-occurring terms that highlight emerging thematic patterns and structures. This choice directly aligns with our research objective to trace the thematic evolution and conceptual convergence towards CoPSS design. Publication-related analysis supplemented this by providing insights into the quantity and distribution of publications over time, allowing for the charting of growth patterns and overall research productivity. Alternative techniques, such as co-citation analysis, were considered but excluded as they primarily highlight historically significant literature and were thus less suited to our objective of identifying emerging conceptual trends.

To address RQ2, thematic analysis was employed to examine the current state of CoPSS design research, including dominant research themes, methodological approaches, theoretical foundations, and proposed research directions as identified in the literature. This method identified recurring themes within the textual data, directly emerging from the literature content. The analysis facilitated categorisation and interpretation, enabling a nuanced understanding of current research practices and perspectives in CoPSS design (Rodríguez et al., 2020).

To address RQ3, an integrative framework was constructed based on the synthesis of bibliometric and thematic findings. This framework captures the intersection of social, technical, and ecological dimensions in CoPSS design, mapping areas of thematic convergence, methodological diversity, and structural gaps. It draws directly from the analysed literature to highlight where integration occurs and where fragmentation persists. In doing so, it also identifies gap-driven research themes that reflect unmet needs and emerging directions for future inquiry, providing a structured foundation for guiding research on holistic and adaptive CoPSS design.

2.4. Results presentation and synthesis

The findings are structured to sequentially address the research questions, with visual representations included for enhanced clarity and interpretability. Section 3 presents bibliometric findings addressing RQ1 through a keyword co-occurrence map (using VOSviewer) and statistical charts illustrating publication trends. Section 4 addresses RQ2, presenting thematic findings regarding research themes, methods, adopted theoretical lenses, and proposed future directions, visualised through various diagrams. Venn diagrams were used to map the distribution of research themes and future directions across social, technical, and ecological dimensions, revealing gaps in integration. Tree diagrams were used to illustrate the distribution of research methods across CoPSS design literature and to provide a structured typology of theoretical lenses, highlighting the diversity and clustering of theoretical perspectives in CoPSS design research. Finally, Section 5 introduces an integrative framework (RQ3) that synthesises these findings into a coherent visual representation of CoPSS design research. This framework serves as both a synthesis of state-of-the-art knowledge and a roadmap to guide future research. These visual formats were chosen to enhance interpretability and to support the identification of theory-informed, gap-driven insights in CoPSS design research.

3. Trends toward CoPSS design amid the triple transitions

This section presents the findings from a bibliometric analysis of the 723 screened PSS design articles, including a keyword co-occurrence mapping that reveals structural proximity between author keywords, as well as an analysis of publication totals and trends. This analysis aims to reveal the trend toward CoPSS design within the broader context of the PSS design literature.

3.1. Keyword co-occurrence mapping of PSS design literature

Fig. 2 illustrates the keyword occurrences network derived from 723 initial screened articles, using VOSviewer to visualise author keyword relationships. To ensure interpretability, a minimum threshold of five keyword occurrences was applied to 2126 author keywords, producing a manageable set of 60 keywords displayed (Eck and Waltman, 2009). A harmonised thesaurus was employed to merge synonyms and unify terminology across diverse author styles (e.g., "product service system" and "product-service-system").

The resulting map reveals three thematic clusters, differentiated by colour and conceptual orientation: servitisation (red), sustainability (green), and digitalisation (blue). The size of the nodes within each cluster indicates the frequency of keyword occurrences. Edge thickness denotes Total Link Strength (TLS), which is calculated in VOSviewer as the sum of a keyword's co-occurrence link weights with all other keywords, indicating how strongly that term is connected across the network. The layout uses VOSviewer's Lin-Log normalisation, which positions nodes proportionally to their link strength while maximising cluster modularity, thus sharpening thematic boundaries and reducing visual clutter (Eck and Waltman, 2009). The inset tables list, for each cluster, the keywords along with their Occurrence (Occ.) and TLS values, accompanied by an exemplar article to maintain transparency in qualitative anchoring (Snyder, 2019; Tranfield et al., 2003; Zupic and Čater, 2015).

3.1.1. Cluster 1: servitisation-dominant cluster

Cluster 1 contains 30 high-frequency keywords, led by "product-service system" (Occ. 439, TLS 462), "servitisation" (Occ. 64, TLS 107), "service design" (Occ. 21, TLS 41), and "business modelling" (Occ. 21, TLS 38). These nodes collectively attract a large number of co-occurrence edges, accounting for a substantial portion of the network's total weight. This provides evidence that the servitisation discourse continues to provide the structural core of PSS design

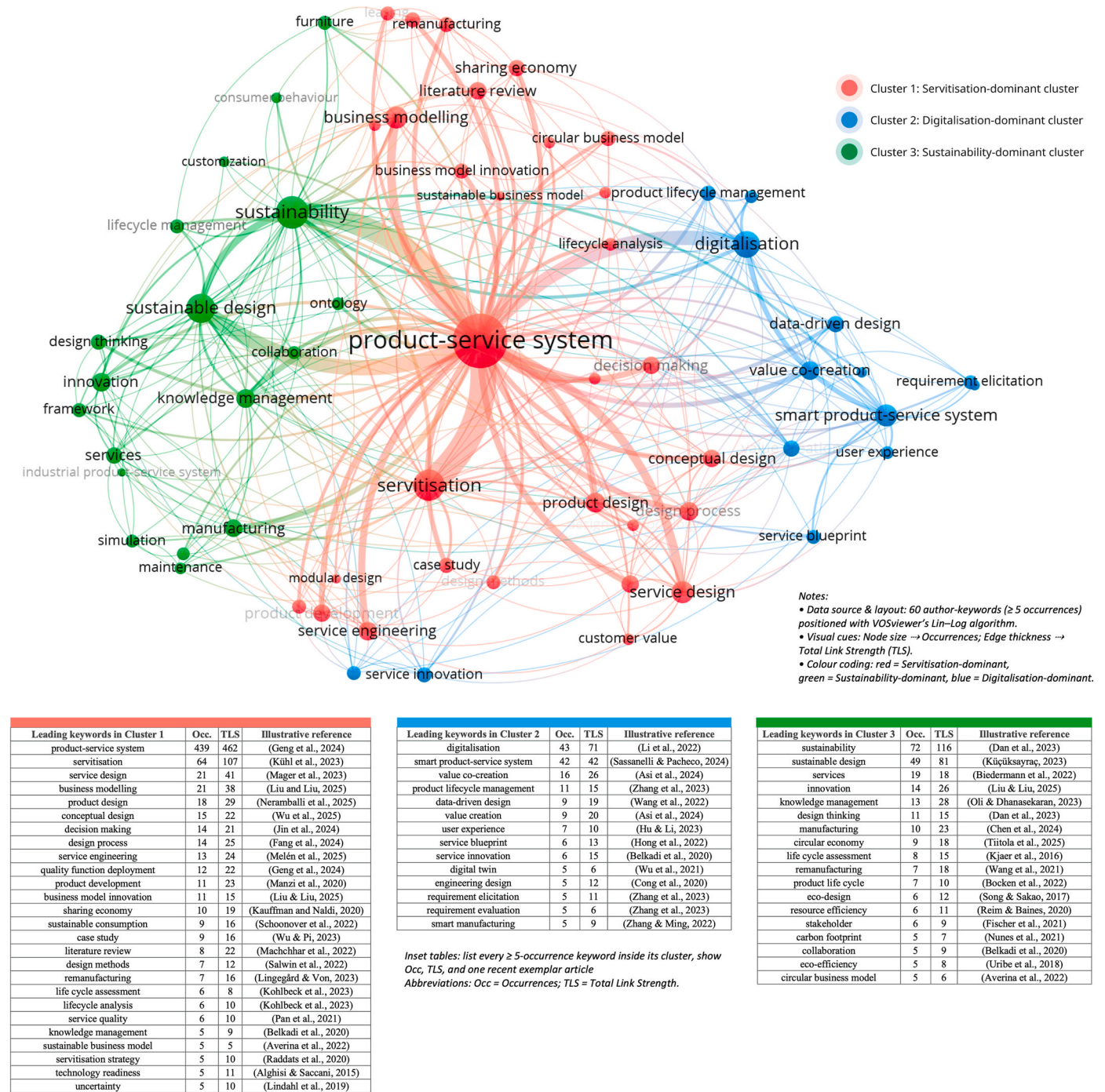


Fig. 2. Keyword occurrences analysis of PSS design literature (analysed using VOSviewer).

research, marking a transition from product-oriented to service-integrated logic (Tukker, 2004). It also indicates that this cluster remains strongly aligned with economic and organisational design logics. For example, these high-occurrence and strongly linked terms co-occur with design process-oriented terms, including "design process" (Occ. 14, TLS 25) and "product design" (Occ. 18, TLS 29).

These high-frequency, strongly connected, and thematically coherent keywords indicate a high degree of conceptual maturity within this cluster. This is reflected in the methodological sophistication of representative studies, which develop and operationalise service design frameworks, modelling tools, and value-oriented strategies. For instance, Pereira et al. (2019) analysed the role of network configurations in enabling value co-creation across service-led innovation.

Annarelli et al. (2020) highlighted how PSS enables lifecycle cost optimisation and stakeholder integration. A focus on design integration, particularly regarding how service elements are embedded into product-centric models, is evident in both conceptual and empirical studies (Sassanelli et al., 2018).

Other prominent keywords, such as "product design" (Occ. 18, TLS 29), "design process" (Occ. 14, TLS 25), and "quality function deployment" (Occ. 12, TLS 22), indicate the centrality of system-level design and the translation of customer requirements into service-enhanced solutions. Notably, "conceptual design" (Occ. 15, TLS 22) and "decision making" (Occ. 14, TLS 21) highlight a growing emphasis on early-stage planning and trade-off negotiation (Bertoni and Bertoni, 2020; Geng et al., 2010).

Interestingly, sustainability-related keywords such as "remanufacturing" (Occ. 7, TLS 16) and "sustainable consumption" (Occ. 9, TLS 16) also appear in this cluster, although they remain peripheral. Their presence may indicate initial efforts to embed sustainability considerations into servitisation strategies, aligning with the broader trend of early-stage integration toward CoPSS. However, their low frequency and limited connectivity reflect that such convergence is still in its infancy and not yet structurally embedded within the cluster's thematic core.

3.1.2. Cluster 2: digitalisation-dominant cluster

Cluster 2 contains 14 keywords with a clear focus on digital integration into PSS design. It is anchored by "digitalisation" (Occ. 43, TLS 71), "smart product-service system" (Occ. 42, TLS 42), and "value co-creation" (Occ. 16, TLS 26). These high-TLS terms illustrate a growing discourse on how digital infrastructures reshape PSS design, particularly through real-time data, platformisation, and feedback mechanisms (Zhang et al., 2023). Keywords such as "digital twin" (Occ. 5, TLS 6), "data-driven design" (Occ. 9, TLS 19), and "user experience" (Occ. 7, TLS 10) reflect increasing attention to information architectures and design intelligence that support system responsiveness and personalisation (Wu et al., 2021).

The structural positioning of "product lifecycle management" (Occ. 11, TLS 15), "requirement elicitation" (Occ. 5, TLS 11), and "engineering design" (Occ. 5, TLS 12) highlights a convergence between engineering methods and digital service development. These linkages suggest the emergence of a computationally enabled and lifecycle-informed design paradigm (Pirola et al., 2020).

Despite the relative recency of this cluster's emergence (most terms peak after 2020), it exhibits high connectivity and conceptual density, indicating rapid development. The presence of "service innovation" (Occ. 6, TLS 15), often associated with servitisation, suggests a conceptual crossover now being recontextualised within digital design environments (Belkadi et al., 2020).

This cluster portrays a catalytic role in enabling system-level integration and real-time adaptation in CoPSS design. It reflects a strong thematic emphasis on digital infrastructure, user interaction, and system responsiveness. While its overall connections to the other clusters are limited, the notable exception is the strong co-occurrence link between "digitalisation" and "product-service system", inherited from the servitisation cluster, which anchors the digitalisation discourse within the broader logic of PSS. This tie illustrates a social-technical orientation, where digital tools support service integration and adaptive capabilities. Nevertheless, links to ecological and sustainability-oriented concepts are weak, indicating that while integration across the triple transitions exists, it remains partial and underdeveloped in the digitalisation cluster.

3.1.3. Cluster 3: sustainability-dominant cluster

The sustainability-dominant cluster is anchored by "sustainability" (Occ. 72, TLS 116) and "sustainable design" (Occ. 49, TLS 81), indicating strong internal coherence focused on environmental objectives and design-driven strategies. These terms co-occur frequently with "innovation" (Occ. 14, TLS 26) and "knowledge management" (Occ. 13, TLS 28), suggesting an orientation toward organisational learning and capability development for sustainable outcomes.

Of particular significance is the appearance of "services" (Occ. 19, TLS 18) and "industrial product-service system" (Occ. 5, TLS 3), which signal a potential convergence with the servitisation discourse. However, their weak connectivity and peripheral positions indicate that these links remain underdeveloped. Likewise, digital-related terms are largely absent or poorly connected, reflecting a lack of integration with the digitalisation cluster.

Despite this, some signals of convergence are emerging. The presence of "design thinking" (Occ. 11, TLS 15), "simulation" (Occ. 5, TLS 10), and "collaboration" (Occ. 7, TLS 9) reflects a gradual shift toward systemic and participatory approaches that are compatible with CoPSS logic

(Afshar and Wang, 2010). These keywords suggest that sustainability research is beginning to engage with the social and technical dimensions of design, even if full integration remains limited.

The sustainability cluster exhibits a well-established focus on environmental and design strategies, while also demonstrating initial, albeit limited, connections to service and digitalisation themes. The presence of peripheral terms, such as "services" and design-related concepts, signals an early movement toward integration. Although these links remain weak, they reflect a nascent trend toward convergence across the triple transitions.

3.2. Publication trends across the triple transitions

Fig. 3 presents a categorical breakdown of the 723 reviewed articles based on the analysis of total publication numbers (Donthu et al., 2021) of research related to each transition and their intersections. Literature discusses at least one transition predominantly in isolation: servitisation (14 %), digitalisation (11 %), and sustainability (28 %). Studies combining two transitions are far fewer, 4 % on sustainability and servitisation, or sustainability and digitalisation, and 3 % on servitisation and digitalisation, while only 1 % of papers encompass all three transitions, indicating a nascent but growing interest in studying the interplay between these transitions in PSS literature. In contrast, 35 % of the articles did not match any of the transition-related search strings and were therefore categorised as "Other." These studies may focus on general PSS design issues or adopt alternative framings that do not explicitly employ the selected keywords.

Chronologically, Fig. 4 presents the analysis of annual publication trends from 2003 to 2024, showcasing marked growth in studies concerning each of the triple transitions. Fig. 4 (1) shows a steady growth in publications addressing each transition, with a sharp rise in sustainability-focused PSS design after 2015, coinciding with the proliferation of sustainability reporting standards and the rise of circular economy policy frameworks. Servitisation- and digitalisation-focused studies exhibit slower but steady growth, reflecting an ongoing shift in industrial priorities towards service innovation and digital transformation. Notably, Fig. 4 (2) highlights the emergence of twin-transition and triple-transition studies, which have only begun to appear with noticeable frequency in the past five years. This recent uptick suggests a shifting research landscape where service, sustainability, and digital concerns are increasingly examined in tandem, albeit still in a limited and fragmented manner.

Overall, this section contributes a structured understanding of how CoPSS design research has evolved amid the triple transitions by combining two complementary perspectives. The keyword co-occurrence analysis and transition category pie chart (Figs. 2 and 3) reveal the horizontal thematic structure of the PSS design field, showing how servitisation, digitalisation, and sustainability are currently positioned in relation to one another. In contrast, the publication trend

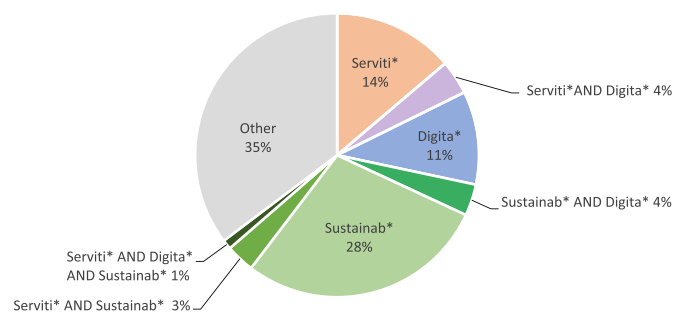


Fig. 3. PSS design documentation related to the triple transitions. Notes: The asterisk (*) used in the strings replaces multiple characters anywhere in a word. For example, behav* finds behave, behavior, behaviour, behavioural, behaviourism, etc.

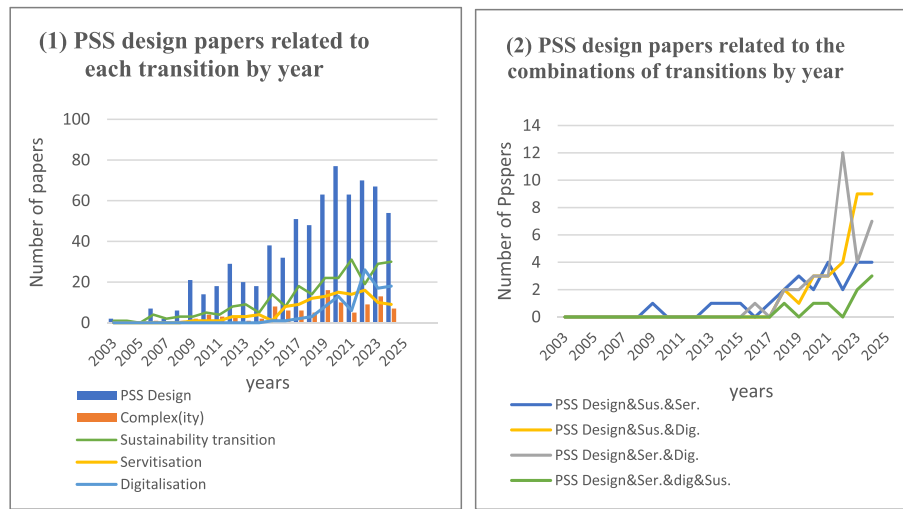


Fig. 4. PSS design papers related to the combinations of triple transitions by year.

analysis (Fig. 4) provides a longitudinal (vertical) perspective, illustrating how research attention to each transition and their combinations has developed over time. Together, these findings expose an uneven and

fragmented engagement with the triple transitions. Most contributions still approach servitisation, sustainability, and digitalisation as distinct lines of inquiry, with limited cross-pollination. The scarcity of triple-

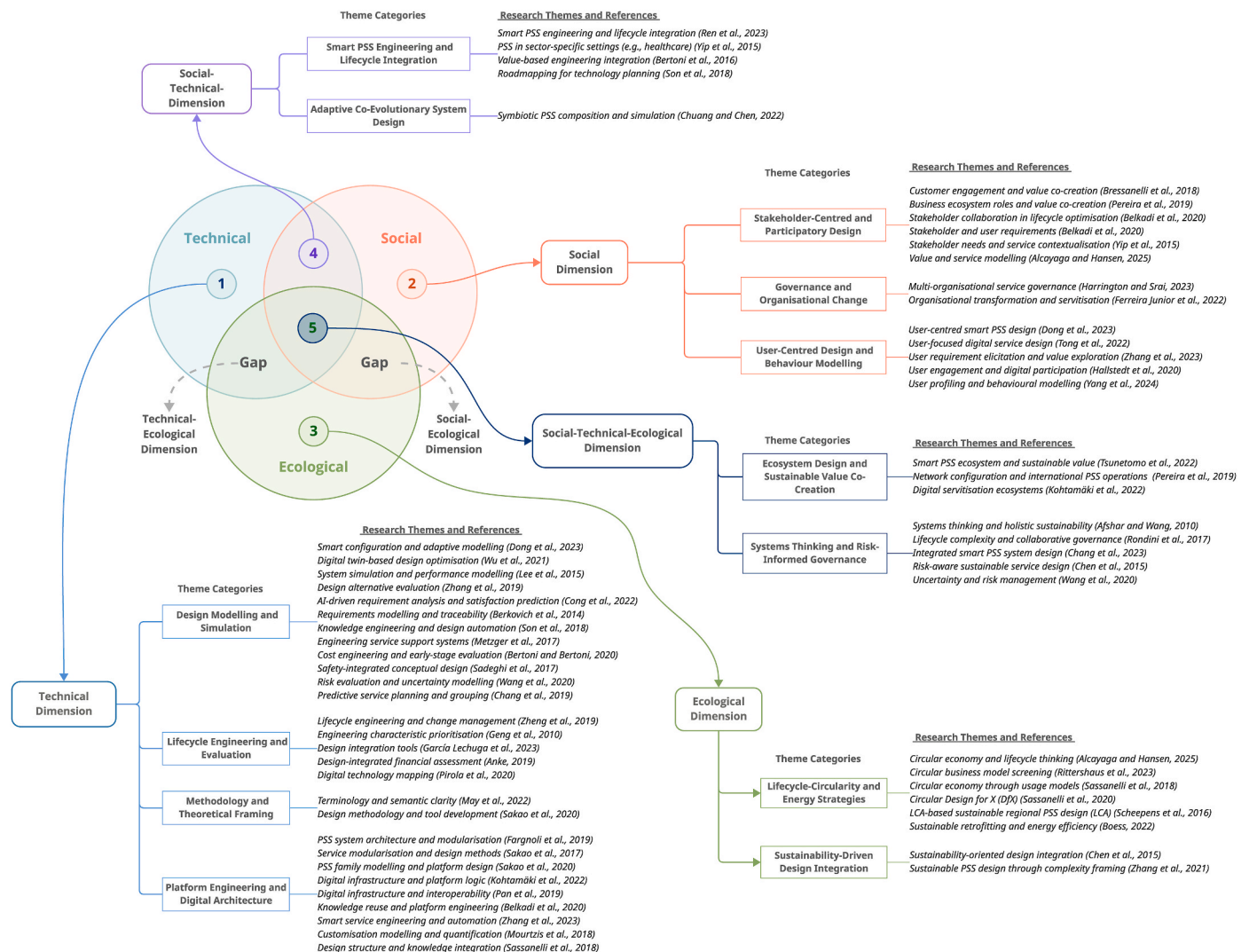


Fig. 5. Dominant Research themes in CoPSS Design literature.

transition research underscores a missed opportunity for developing more integrated, systemic approaches to CoPSS design.

It is important to clarify that the analyses in this Section 3 are based on keyword-driven search strings applied to article titles, abstracts, and keyword fields. As such, the categorisations reflect the explicit use of transition-related terminology rather than the conceptual depth of integration. This approach may underrepresent studies that implicitly engage with these transitions or use alternative vocabularies. While this method offers a broad overview of publication patterns and thematic clusters, it does not fully capture the complexity or nuance of transition integration. To mitigate this limitation, the next section presents a fine-grained full-text thematic analysis, which examines CoPSS design research more holistically, identifying conceptual convergence and structural gaps across social, technical, and ecological dimensions.

4. Current landscape of CoPSS design research

This section aims to delve into RQ2 about how CoPSS design has been studied. We present the findings from four perspectives: dominant research themes, employed methodological approaches, mobilised theoretical foundations, and literature-proposed research directions.

4.1. Dominant research themes

The research focus of the selected papers was coded into themes, which were then grouped into aggregated theme categories based on similar research contributions. Fig. 5 synthesises the research themes of the final 62 CoPSS design articles, categorised across the technical, social, ecological, and intersecting domains. As this study argues, CoPSS is conceptualised as a social–technical–ecological system shaped by the intertwined transitions of servitisation, digitalisation, and sustainability. This mapping allows a systematic assessment of how current research addresses these multiple dimensions and where integration remains limited.

Across the literature, the technical domain emerges with the greatest concentration of research effort. A substantial number of studies focus on modelling, simulation, digital infrastructure, and system integration. These themes reflect a continued emphasis on advancing the analytical and computational capabilities needed to design and deliver CoPSS. For instance, Dong et al. (2023) develop smart configuration models to support adaptive design; Wu et al. (2021) leverage digital twins to optimise performance parameters; and Cong et al. (2022) apply AI to predict requirement satisfaction. Additional contributions by García Lechuga et al. (2023) and Zhang et al. (2023) focus on design integration tools and service automation, while Kohtamäki et al. (2022) examine digital platform logic. These studies often aim to ensure scalability, efficiency, and reliability under increasing design complexity. However, most studies in this domain prioritise system functionality and performance, with limited discussion of how technical architectures can enable or constrain stakeholder engagement or sustainability goals. This reflects a broader pattern noted in PSS engineering research, where technical excellence has historically been prioritised over system-wide sustainability trade-offs or participatory governance mechanisms (Rodríguez et al., 2020).

The social domain has seen thematic expansion across four categories: co-creation and engagement, governance and organisational change, and user-centred behavioural modelling. These categories reflect a shift toward actor-centred CoPSS design, in which stakeholder interaction and behavioural insights are essential to service configuration. For instance, Harrington and Srai (2023) examine multi-actor governance models, and Ferreira Junior et al. (2022) highlight organisational adaptation in response to servitisation. Hallstedt et al. (2020) and Yang et al. (2024) develop digital engagement strategies and behavioural profiling to align service delivery with user preferences. While these studies advance the understanding of actor coordination, participatory sensemaking, and behavioural adaptation, they often lack

coordination with system-level modelling or enterprise design, as noted by Bertoni et al. (2016) and Kohtamäki et al. (2022). The emphasis on participatory processes and behavioural insights tends to produce locally valid design knowledge. Studies such as Ferreira Junior et al. (2022) remain focused on specific organisational transitions or cases, raising the question of how these insights scale or interoperate with complex infrastructure and multi-actor systems. This lack of operational generalisability limits the influence of social design contributions when design moves from project-specific initiatives to integrated CoPSS platforms.

The ecological domain features a more focused set of studies. These are primarily concerned with lifecycle thinking, circular design, and sustainability integration. Alcayaga and Hansen (2025) and Rittershaus et al. (2023) introduce tools for circular business model screening and service-value modelling. Sassanelli et al. (2020) develop Design for X frameworks that embed circularity principles into technical design. These contributions reflect a growing maturity in sustainability-oriented design methods, but they tend to operate independently of digital architecture or stakeholder co-design. This thematic isolation may stem from the continued separation of environmental design mandates from core service design and collaboration processes, as previously noted by Vezzoli et al. (2015).

The social–technical cluster is more established. It includes studies focused on smart PSS engineering, service co-evolution, and technology roadmapping, where digital capabilities and stakeholder coordination intersect to support service adaptability (Chuang and Chen, 2022; Ren et al., 2023; Son et al., 2018). These contributions highlight how technical systems are increasingly configured in ways that respond to user participation and organisational dynamics.

Notably, the social–ecological and technical–ecological intersections remain absent as distinct categories in the thematic structure. This absence may reflect that environmental factors are often treated as exogenous to engineering-driven optimisation models or stakeholder-oriented governance frameworks. Instead of being embedded within participatory tools or digital infrastructures, they are typically addressed through parallel or post hoc evaluation. This structural asymmetry indicates a continued fragmentation in how CoPSS design research approaches systemic sustainability.

Despite the absence of explicit social–ecological or technical–ecological categories, the emergence of social–technical–ecological categories suggests that integration tends to occur only when all three dimensions are explicitly addressed. One explanation may be that research which begins from a combined social–technical lens is more likely to acknowledge ecological impact in the context of systemic coordination, particularly where ecosystem governance and lifecycle value are foregrounded. In contrast, studies that focus solely on social or technical entry points may lack the frameworks, mandates, or empirical settings necessary to expand into environmental considerations. For instance, Tsunetomo et al. (2022) and Chang et al. (2023) explore ecosystem-level design frameworks that combine stakeholder coordination, lifecycle management, and sustainable value generation. Wang et al. (2020) incorporate ecological uncertainty into service configuration models. These works reflect promising integration efforts that address the multi-dimensional challenges of CoPSS. However, these contributions remain limited in number and often conceptual.

In summary, the current dominant research themes of CoPSS design research demonstrates strong domain-specific maturity in both technical and social approaches, as well as increasing methodological innovation in ecological design. However, it also reveals a notable absence of social–ecological and technical–ecological integration themes. While social–technical integration is more visible, and social–technical–ecological contributions are emerging, the integration remains fragmented. A critical challenge lies in developing design strategies that link stakeholder engagement and technical performance evaluation with environmental imperatives to enable a more coherent and systemic approach to CoPSS design.

4.2. Employed methodological approaches

Fig. 6 presents the methodological landscape of the reviewed CoPSS design papers.

The most employed approaches are empirical case-based studies, typically conducted in organisational settings such as manufacturing firms (Afshar and Wang, 2010), healthcare systems (Yip et al., 2015), and energy infrastructure projects (Chang et al., 2023). These studies have been instrumental in testing and refining existing frameworks, such as the assessment of organisational readiness for transformation (Ferreira Junior et al., 2022) or analysis of service architectures in the healthcare sector (Yip et al., 2015). However, most contributions in this category are explanatory and deductive, with limited use of inductive or longitudinal case designs that could uncover novel mechanisms or support grounded theory development in emerging CoPSS contexts. This indicates a methodological limitation in revealing how CoPSS initiatives emerged from real-world contexts or evolved under shifting operational conditions. In particular, the limited use of inductive methods weakens the field's ability to theorise from practice and to capture the dynamic and situated nature of CoPSS development as it unfolds in diverse empirical contexts.

A second, rapidly expanding category is decision analysis and multi-criteria methods, which have become prominent in supporting design choices involving trade-offs among economic, social, and environmental objectives. Methods such as entropy-weighted Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) (Zheng et al., 2019), fuzzy Analytic Hierarchy Process (AHP) (Chen et al., 2015), and variable-precision rough sets (Chang et al., 2023) are frequently employed to structure evaluation problems and prioritise design alternatives. These methods introduce analytical depth and rigour, particularly in modelling stakeholder preferences and sustainability criteria. However, they are often used in a static, post-hoc manner rather than embedded in real-time or iterative decision environments. Although

several studies acknowledge the potential synergy between decision-support tools and simulation-based approaches (Bertoni and Bertoni, 2020; Mourtzis et al., 2018b), such integration remains underdeveloped. Few papers demonstrate actual empirical implementation of dynamic coupling with digital twins or adaptive feedback loops in CoPSS contexts (Wu et al., 2021; Zhang et al., 2023); such integrations remain rare and insufficiently developed to establish a consistent methodological pattern.

The families of design methods and engineering structures and design-oriented conceptual modelling strategies represent a complementary methodological stream aimed at structuring and theorising the design process itself. Studies in this space use protocol analysis (Shimomura et al., 2015), industrial pilot validations (Sakao et al., 2020), and design structure matrices to trace the logic and flow of design activities. Others propose structured conceptual artefacts such as user profiling systems (Yang et al., 2024) or social-technical integration approaches (Boess, 2022). At the same time, conceptual work has flourished through typology development (Zhang et al., 2021), ontology formalisation (May et al., 2022), and abductively derived service frameworks (Alcayaga and Hansen, 2025). These studies are crucial for advancing theoretical clarity and coherence. However, without efforts to systematically compare or consolidate these conceptual models, such as through synthesis-oriented reviews or comparative evaluations, there is a risk of conceptual fragmentation that hinders cumulative theory-building in CoPSS design.

Although fewer in number, literature-based and discourse analyses play a crucial integrative role by consolidating dispersed knowledge and identifying thematic shifts within the field. For example, Pirola et al. (2020) map the evolution of digital technology roles in PSS, while Lugnet et al. (2020) explore the discursive development of design perspectives. These contributions support theoretical integration across disciplines but remain relatively scarce given the conceptual diversity of CoPSS design research. This underutilisation of literature synthesis

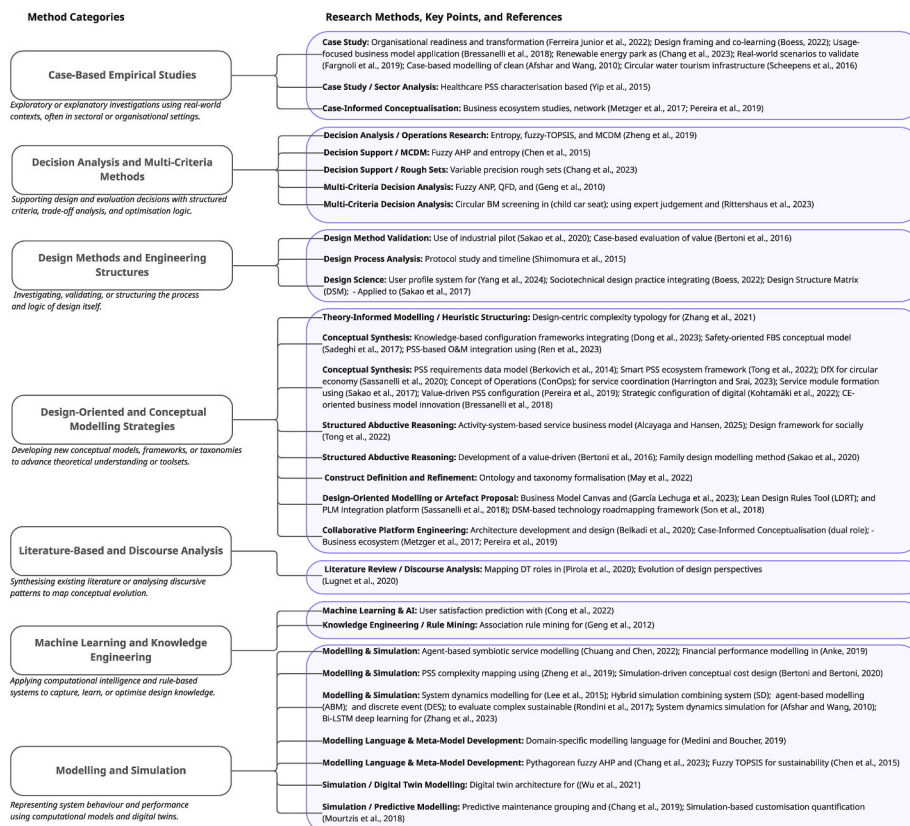


Fig. 6. Employed methodological approaches in CoPSS design literature.

hinders cumulative knowledge building and the identification of boundary conditions for emerging models.

The application of machine learning and knowledge engineering techniques remains exploratory. Supervised learning models have been applied to predict user satisfaction (Cong et al., 2022), while association rule mining has been used to extract design heuristics from datasets (Geng et al., 2012). These early applications indicate promising directions, but wider adoption appears constrained. While several studies acknowledge barriers such as data availability and model transparency (Cong et al., 2022; Geng et al., 2012), there is a lack of critical examination of these issues within the reviewed literature. Although interest in operational data integration and explainable Artificial Intelligence (AI) is increasing, this methodological direction remains underexplored and insufficiently theorised, indicating a gap in both empirical application and conceptual development.

The category of modelling and simulation has become methodologically advanced, particularly through the adoption of hybrid techniques. Traditional system dynamics modelling has been extended through combinations with agent-based modelling and discrete-event simulation

to represent multi-level feedback mechanisms in complex systems (Rondini et al., 2017). Recent developments include agent-based models for symbiotic services (Chuang and Chen, 2022), financial performance simulators (Anke, 2019), and initial applications of deep learning architectures for dynamic modelling (Zhang et al., 2023). Despite their technical sophistication, most of these models operate in isolation from empirical case studies or participatory design processes (Bertoni and Bertoni, 2020; Mourtzis et al., 2018b). Very few studies demonstrate how simulation outputs inform real-time decision-making or are validated through stakeholder engagement (Chuang and Chen, 2022; Zhang et al., 2023), which limits their contribution to actionable system-level design governance.

Overall, the methodological trajectory depicted in Fig. 6 suggests that while empirical case studies continue to dominate, the field is gradually incorporating a broader range of methodological approaches. The increasing use of decision-analytic tools, structured design science methods, and hybrid simulation techniques reflects a growing attempt to align methodological choices with the interdisciplinary, dynamic, and systemic nature of CoPSS design, particularly through the integration of

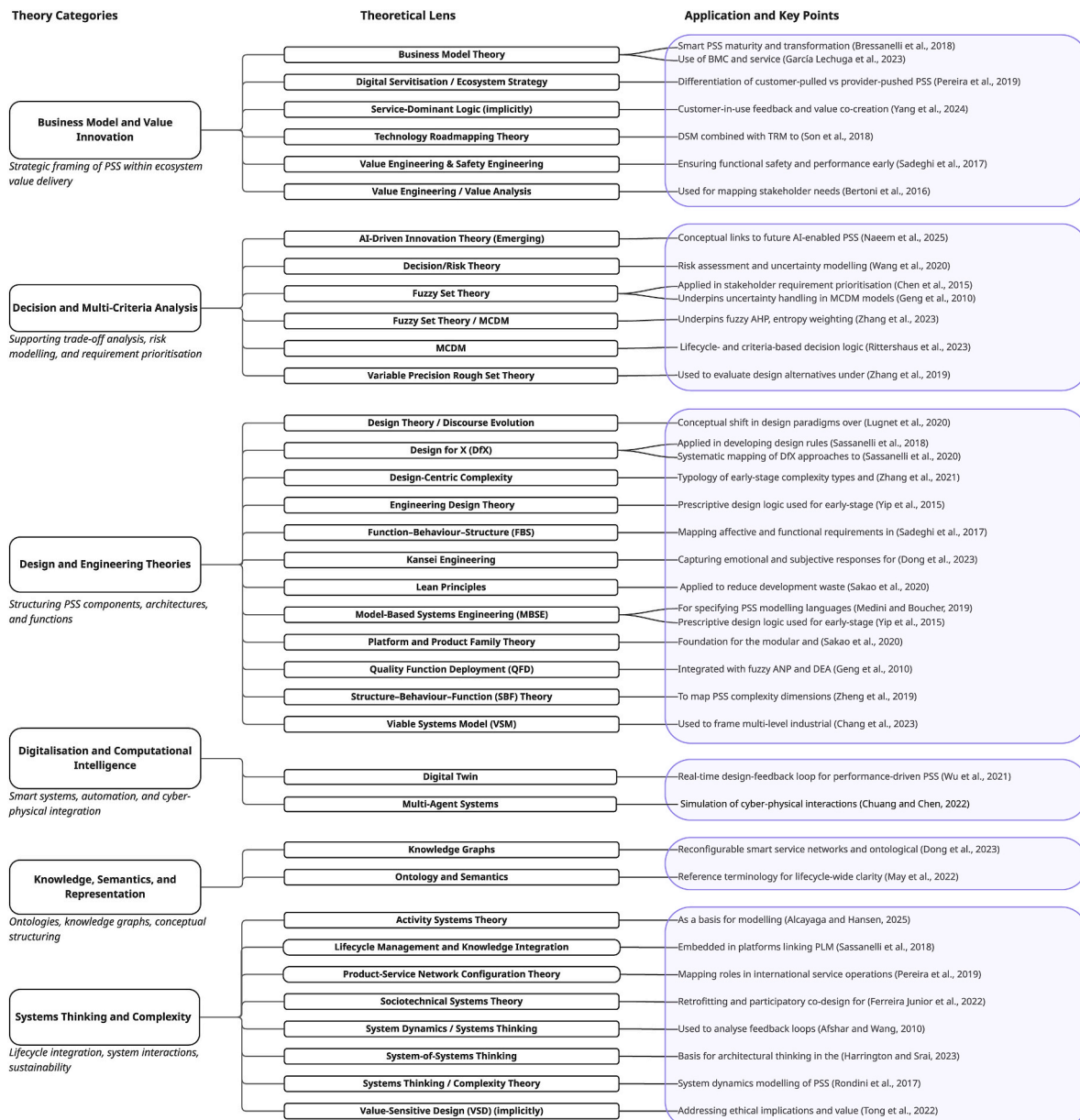


Fig. 7. Mobilised theoretical foundations in CoPSS design literature.

evaluation mechanisms (Chen et al., 2015; Zhang et al., 2019), design logic modelling (Sakao et al., 2020), and computational experimentation (Rondini et al., 2017; Zhang et al., 2023). However, the limited application of inductive, longitudinal research methods (Alcayaga and Hansen, 2025; Boess, 2022) and the underutilisation of real-time adaptive approaches (Mourtzis et al., 2018a; Wu et al., 2021) continue to constrain the development of CoPSS design research's ability to theorise from practice and address the situated complexities. Addressing these methodological limitations is necessary to deepen theoretical insight, enhance responsiveness to changing system conditions, and support the development of empirically grounded, adaptive design strategies for CoPSS.

4.3. Mobilised theoretical foundations

A diverse range of theoretical lenses and analytical perspectives has been used in the reviewed CoPSS design literature to address the complexity of social–technical–ecological systems. These lenses serve distinct purposes: strategic framing, decision support, system structuring, computational integration, knowledge representation, and complexity governance. Fig. 7 presents a structured typology of these foundations.

Business model and value innovation lenses are primarily concerned with strategic framing, focusing on how CoPSS solutions are conceptualised to deliver and capture value. These lenses support business model transformation, servitisation strategies, and value co-creation mechanisms. In practice, Bressanelli et al. (2018) emphasise Smart PSS maturity as a trajectory of business model evolution, whereas García Lechuga et al. (2023) operationalise this through visual tools like Business Model Canvas (BMC), exposing a gap between strategic vision and actionable design. Ecosystem strategy perspectives further clarify distinctions between provider-pushed and customer-pulled models (Pereira et al., 2019). Although Technology Roadmapping is intended to support forward-looking planning, it is often used retrospectively as a classification tool, limiting its capacity to guide anticipatory CoPSS design (Son et al., 2018). Service-Dominant Logic, though often implicit, frames value as co-created through use and interaction (Yang et al., 2024). Collectively, these contributions offer useful strategic framings but often remain at a conceptual level. Despite their conceptual contributions, only a limited number of studies provide empirical evidence on how these lenses inform practical design frameworks, collaborative processes, or feedback mechanisms, undermining their potential to support CoPSS design delivery in complex real-world settings (Bressanelli et al., 2018; García Lechuga et al., 2023; Pereira et al., 2019).

Decision-making and multi-criteria lenses support design decisions under uncertainty, often through trade-off analysis, requirement prioritisation, and sustainability evaluations. Fuzzy Set Theory and Multi-Criteria Decision-Making (MCDM) approaches are commonly employed to structure stakeholder preferences and manage decision complexity (Chen et al., 2015; Geng et al., 2010). Tools like fuzzy Analytic Hierarchy Process (AHP) and entropy-based weighting (Zhang et al., 2023) demonstrate technical sophistication but are typically tested in controlled or static settings, divorced from the iterative, multi-actor processes typical of CoPSS design. Variable Precision Rough Set Theory (Zhang et al., 2019) and AI-driven innovation logic (Naeem et al., 2025) signal computational advancements, yet most contributions remain at a conceptual or pilot level, lacking longitudinal validation. Moreover, although some studies address lifecycle-related trade-offs (Rittershaus et al., 2023), they rarely embed such decisions within broader systemic and adaptive feedback environments. This indicates a persistent fragmentation between decision theory development and the organisational realities of CoPSS design projects.

Design and engineering lenses focus on structuring the components, functions, and process logics of CoPSS. These include both established frameworks, such as classical Engineering Design frameworks that focus

on systematic, phase-based development processes (Yip et al., 2015), and more recent paradigms such as Function–Behaviour–Structure (Sadeghi et al., 2017) and Kansei Engineering (Dong et al., 2023), which incorporate user emotion and cognitive feedback. Emerging lenses support modularisation (Sakao et al., 2020), complexity profiling (Zhang et al., 2021), and cross-layer platform integration (Belkadi et al., 2020). Frameworks such as Design for X (DfX) and Model-Based Systems Engineering (MBSE) are leveraged to align sustainability and digital interoperability in early-stage design (Medini and Boucher, 2019; Sasanelli et al., 2018).

While design and engineering lenses provide structured methodologies for CoPSS development, their operational application often lacks adaptability to real-world project settings. Many studies rely on tool-centric validations or laboratory-based demonstrations, offering limited evidence of how these approaches function within interdisciplinary, uncertain, or social–ecologically sensitive environments (Sakao et al., 2020; Yip et al., 2015; Zhang et al., 2021). This reflects a deeper tendency to prioritise technical coherence over collaborative negotiation or systems integration. As a result, structured design approaches remain underutilised when confronted with the emergent and distributed nature of CoPSS delivery.

Digitalisation and computational intelligence lenses integrate automation, cyber-physical feedback, and data-driven optimisation into CoPSS design. Digital twin systems (Wu et al., 2021) enable simulation and real-time performance monitoring, while multi-agent systems facilitate interaction modelling across distributed components (Chuang and Chen, 2022). These frameworks demonstrate technical potential, yet their applications remain narrow in scope, isolated from ecological outcomes, user interaction patterns, or governance mechanisms. The absence of integrated perspectives across digital, social, and ecological dimensions limits their contribution to the holistic design logic that CoPSS demands.

Knowledge and semantics-based lenses aim to improve conceptual clarity, interoperability, and lifecycle alignment in CoPSS. Ontologies and semantic systems (May et al., 2022) provide reference structures for standardisation, while knowledge graphs enable reconfigurable service networks in contexts such as intelligent family medication (Dong et al., 2023). While these contributions offer foundational insights, their empirical reach remains narrow. The systemic deployment of these approaches in real-world CoPSS governance and collaboration processes remains underexplored.

Systems thinking and complexity lenses encompass a range of approaches focused on modelling feedback, coordinating system elements, and embedding social or ethical considerations into design. These lenses are conceptually rich, reflecting efforts to capture both the dynamism and the layered structure of CoPSS environments. Some perspectives focus on the interaction between human actors, technologies, and ethical commitments, such as activity systems theory (Alcayaga and Hansen, 2025), sociotechnical systems theory (Ferreira Junior et al., 2022), and value-sensitive design (Tong et al., 2022), which underscore the participatory and normative aspects of CoPSS design. Others offer structural coordination insights. For instance, the network configuration lens investigates how roles are organised in distributed service systems (Pereira et al., 2019), while system-of-systems thinking and the viable systems model contribute frameworks for aligning distributed operations and maintaining organisational resilience (Chang et al., 2023; Harrington and Srai, 2023). Meanwhile, system dynamics offers tools to analyse causal feedback and long-term behavioural implications (Afshar and Wang, 2010; Rondini et al., 2017), particularly in capturing complexity over time.

Across the theoretical lenses reviewed, despite their conceptual diversity, several limitations recur. Many studies apply these lenses in isolation, focusing on specific facets of CoPSS design without addressing their systemic and cross-dimensional interdependencies. The limited integration across perspectives—such as between systems thinking, engineering approaches, and business modelling—constrains theoretical

development and weakens their capacity to support holistic design logic. In addition, some widely cited lenses, including service-dominant logic and value-sensitive design, are only applied implicitly, undermining methodological transparency. Most notably, the body of work remains largely deductive and tool-driven, with minimal engagement in empirical inquiry that reflects the realities of complex organisational settings. Few studies adopt approaches that derive insights from practice, which limits understanding of how theoretical lenses operate under the dynamic and context-dependent conditions typical of CoPSS design.

4.4. Literature-proposed research directions

This section presents the findings derived from a structured review of proposed research directions across the CoPSS design literature. These proposals, extracted from the reviewed studies, reflect how scholars have identified the limitations of current research and suggested pathways for future inquiry. Fig. 8 illustrates a synthesis of these directions.

A substantial body of literature has pointed to the disconnect between conceptual sustainability ambitions and empirically validated ecological performance in CoPSS design. Scheepens et al. (2016) and Sassanelli et al. (2020) highlighted the challenge of generalising regional Life Cycle Assessment (LCA) results and the predominance of concept-driven circular economy design research with limited real-world grounding. This concern is echoed in studies by Tong et al. (2022) and Bressanelli et al. (2018), which identified the absence of robust lifecycle carbon or resource modelling to substantiate circularity claims. Chen et al. (2015) added that simplified metrics often underpin sustainability evaluations, undermining their practical validity.

Scholars have thus proposed advancing modelling approaches that integrate context-sensitive trade-offs and lifecycle metrics. Rittershaus

et al. (2023) noted the neglect of economic-environmental trade-offs in small-scale PSS, while Zhang et al. (2021) observed that design complexity is seldom linked to long-term sustainability indicators. Contributions from Boess (2022) and Alcayaga and Hansen (2025) further underscored the inconsistent implementation of sustainability tools and the lack of actionable pathways in smart circular economy models. Collectively, these studies call for future work to substantiate sustainability ambitions with context-sensitive, data-informed design frameworks.

The need to address the human and organisational complexity of CoPSS has been widely recognised. Yip et al. (2015) and Pereira et al. (2019) identified that early-stage design often overlooks stakeholder diversity and lacks mechanisms to operationalise co-creation. Tsunetomo et al. (2022) highlighted that value and ethical tensions are underexplored, particularly in digitalised PSS contexts where normative dimensions are increasingly significant.

The literature further points to a limited understanding of organisational learning in longitudinal CoPSS processes. Ferreira Junior et al. (2022) argued that capability-building remains an understudied area, while Dong et al. (2023) noted that affective user feedback is rarely integrated into real-time PSS design, despite its relevance to user engagement. Together, these proposals emphasise the need for methodological development that accounts for evolving stakeholder roles, ethical value negotiations, and user-centred feedback mechanisms.

Many contributions converge in calling for tighter coupling between lifecycle feedback and technical modelling. Belkadi et al. (2020), Sakao et al. (2017), and Geng et al. (2012) each noted that although modular tools and digital models are prevalent, few approaches effectively incorporate real-time lifecycle data. Pan et al. (2019) and Lee et al. (2015) found that the integration of Internet of Things (IoT)

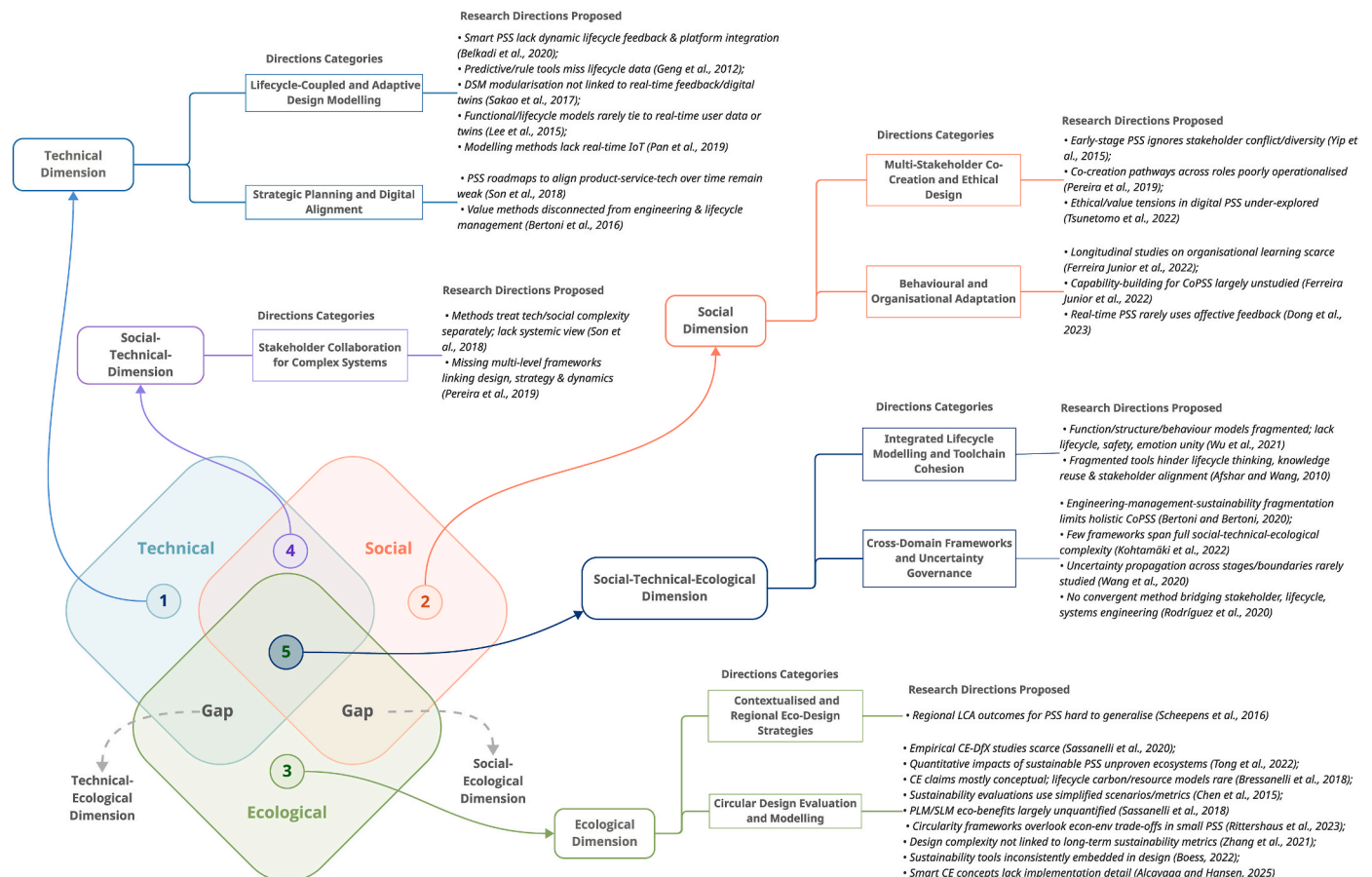


Fig. 8. Proposed research directions in CoPSS design research.

connectivity and digital twins remains limited, thereby constraining system responsiveness.

Strategic misalignment is also identified as a barrier. Bertoni et al. (2016) pointed out that value modelling is often disconnected from lifecycle and engineering functions, while Son et al. (2018) observed weaknesses in roadmapping methods that attempt to synchronise product-service-technology trajectories. These studies propose the development of integrated, platform-oriented design environments that accommodate adaptive system behaviour and align technical modelling with lifecycle realities.

Finally, the literature has consistently highlighted fragmentation in tools and conceptual frameworks as a critical challenge to advancing CoPSS design. Son et al. (2018) and Pereira et al. (2019) observed that social and technical dimensions are typically addressed in silos, and that few frameworks exist to link design, strategic, and organisational levels. Wu et al. (2021) reported the absence of unified models that connect functional, behavioural, and structural elements across safety, emotion, and lifecycle dimensions.

Other studies propose the development of integrative and uncertainty-aware design methods. Afshar and Wang (2010) and Wang et al. (2020) noted that fragmented tools hinder knowledge reuse, lifecycle thinking, and risk propagation analysis. Kohtamäki et al. (2022) critiqued the superficial application of systems thinking in the field, while Rodríguez et al. (2020) called for semantic frameworks that bridge stakeholder perspectives with lifecycle system logic. These contributions

suggest a direction toward convergence—developing ontologies, modelling platforms, and multi-level design logic that enable coherent, transdisciplinary, and socially responsive CoPSS design.

5. Integrative framework for advancing CoPSS design research

To consolidate the findings of this review and guide future inquiry, this section introduces an integrative framework (Fig. 9) that maps the conceptual landscape of CoPSS design research. It brings together the dominant thematic strands addressed by the reviewed literature and the research directions proposed within those studies, thereby positioning research gaps within a structured CoPSS design synthesis. In doing so, the framework reveals both areas of consolidation and persistent fragmentation across social, technical, and ecological dimensions and their intersections. In parallel, by synthesising the methodological approaches employed, and the theoretical foundations mobilised across the reviewed studies, this section also identifies underexplored methodological needs necessary to advance more integrated and reflective CoPSS design research.

Fig. 9 presents a structured framework that synthesises the CoPSS design research landscape across the social, technical, and ecological dimensions and their intersections. The framework is organised into three analytical layers: dominant research themes consolidated from the reviewed literature, research directions proposed within those studies, and gap-driven research themes identified through this review's

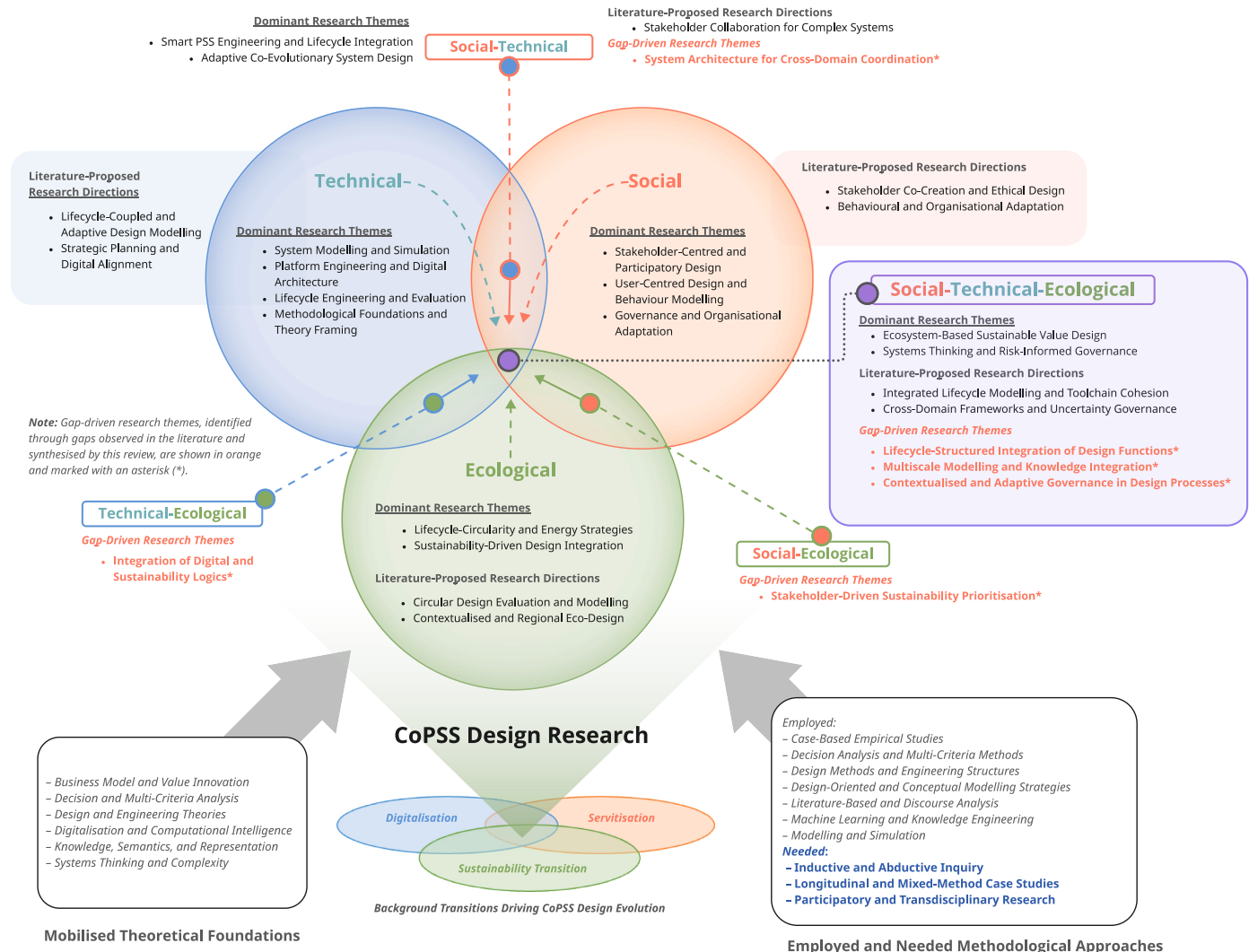


Fig. 9. Integrative framework: the landscape of CoPSS design research.

synthesis of conceptual and methodological limitations. It also incorporates both the theoretical foundations mobilised and the methodological approaches either employed or required, thereby offering an integrated lens through which to understand CoPSS design research. Positioned within the broader context of the sustainability, digitalisation, and servitisation transitions, this holistic structure highlights both areas of convergence and fragmentation in the literature, supporting a more integrated and strategically coherent research agenda.

5.1. Future research directions: towards integrated CoPSS design

Building on the integrative framework, this subsection delineates key directions for future research. These directions correspond to the gap-driven research themes identified in Fig. 9 and reflect integration across specific social–technical–ecological dimensions.

In the social–technical domain, the gap-driven research theme of System Architecture for Cross-Domain Coordination highlights recurring misalignment between stakeholder coordination and design integration across architectural layers. While several studies address behaviour change, lifecycle design, and systems engineering (Kohtamäki et al., 2022), they typically examine these components in isolation. This fragmentation hinders the development of cohesive CoPSS design frameworks capable of managing organisational complexity. Addressing this requires research on stakeholder-configured design structures, particularly under conditions of conflict, distributed accountability, and dynamic project phasing.

In the technical–ecological dimension, the gap-driven research theme of integrating digital and sustainability logics underscores an overemphasis on short-term optimisation at the expense of long-term sustainability responsiveness (Wang et al., 2020; Zhang et al., 2021). Digital infrastructures and modelling tools have rarely been positioned to address carbon impacts, adaptive system behaviours, or ecological feedback mechanisms. Research could advance integrated platforms that embed sustainability logics into their computational architectures and control strategies, offering a more reflexive and ecologically responsive approach to CoPSS engineering.

The social–ecological dimension is reflected in the gap-driven theme of Stakeholder-Driven Sustainability Prioritisation. While participatory methods are prevalent in design-oriented work (Bertoni and Bertoni, 2020; Boess, 2022), there is a limited understanding of how actors reconcile divergent values, institutional interests, and resource constraints in establishing sustainability agendas. Future research is recommended to examine deliberative tools, design negotiation processes, and governance models that account for power asymmetries and inclusivity in priority setting.

At the intersection of social, technical, and ecological aspects, this review identifies three interrelated research directions. First, lifecycle-structured integration of design functions is encouraged to move beyond prevailing compartmentalised approaches, where sustainability objectives are isolated within specific phases or relegated to end-of-life considerations (Afshar and Wang, 2010; Wu et al., 2021). Future studies should investigate how lifecycle architectures can coherently embed environmental and social goals throughout all design stages, enabling sustained alignment across engineering decisions, stakeholder expectations, and policy frameworks.

Second, multiscale modelling and knowledge integration are crucial for addressing persistent misalignments across cognitive, technical, and organisational scales. Existing frameworks, as noted by Kohtamäki et al. (2022) and Rodríguez et al. (2020), often fail to enable vertical coherence between system-level simulation and institutional realities or to support horizontal collaboration across disciplinary silos. Future research should prioritise integrated modelling platforms capable of synchronising quantitative simulations with stakeholder value framings, facilitating iterative negotiation and adaptation across domains.

Third, the role of contextualised and adaptive governance warrants deeper exploration. Although systems thinking has informed high-level

conceptual models, there remains a lack of empirical insight into how governance structures evolve in response to shifting social–technical conditions and ecological imperatives (Wang et al., 2020). Future investigations should focus on governance mechanisms that can operate reflexively within dynamic design environments, accommodating shifting actor networks, regulatory volatility, and diverse interpretations of sustainability. Advancing such governance frameworks will be crucial in enabling resilient and ethically grounded CoPSS strategies that are responsive to real-world complexities.

5.2. Methodological and theoretical advancement

By examining the methodological approaches and theoretical foundations in the reviewed literature, the integrative framework underscores the need to diversify research methods and to combine theoretical lenses to advance CoPSS design research.

Regarding employed methodological approaches, CoPSS research continues to rely heavily on deductive approaches, particularly those using optimisation tools, simulation models, and computational analysis (Afshar and Wang, 2010; Ferreira Junior et al., 2022). While these approaches are useful for evaluating technical efficiency or testing predefined system models, they are limited in their ability to generate new theoretical insights. Specifically, they offer little capacity to reveal the underlying mechanisms through which CoPSS design practices emerge and adapt. Methodological approaches that support theory building, particularly inductive and abductive inquiry, are therefore needed. These can surface explanatory patterns about how design priorities evolve and how social, technical, and ecological considerations are negotiated over time (Boess, 2022).

Despite their potential, participatory and transdisciplinary research approaches remain marginal in CoPSS studies, which tend to prioritise system-centric over stakeholder-driven logics (Biedermann et al., 2022). This gap limits the integration of diverse perspectives, particularly across social and ecological priorities. To address this, participatory and transdisciplinary approaches should be more broadly mobilised to enable collaborative framing of design goals and ensure alignment across stakeholder values and sustainability objectives. Similarly, longitudinal and mixed-method case studies, though seldom employed, are critical for tracing design evolution and behavioural dynamics across project lifecycles and organisational layers (Bertoni and Bertoni, 2020; Kohtamäki et al., 2022). Such methods provide the depth necessary to understand how governance shifts, technological changes, and contextual contingencies co-shape CoPSS outcomes over time.

In terms of theoretical foundations, the literature has mobilised useful lenses, such as business model innovation (Annarelli et al., 2020), systems thinking (Afshar and Wang, 2010), and multi-criteria decision analysis (Zhang et al., 2019). These have often been deployed independently rather than in combination. This fragmented application hinders the field's ability to address the interdisciplinary nature of CoPSS design. Given the inherent integration of social, technical, and ecological systems in CoPSS contexts, there is a pressing need for future research to adopt integrative theoretical lenses. Combining complementary lenses may enable more coherent design logic, facilitate cross-domain knowledge transfer, and enhance the explanatory power of research. Such integrative foundations are better suited to capturing the multifaceted, systemic nature of CoPSS design challenges.

6. Discussion and conclusion

This review examined how the design of Complex Product–Service Systems (CoPSS) has evolved in response to the interrelated transitions of servitisation, digitalisation, and sustainability. It employed a three-stage integrative literature review methodology. First, a bibliometric analysis of 723 Product–Service Systems (PSS) design articles, utilising keyword co-occurrence and publication-related data, identified the structural convergence of servitisation, digitalisation, and sustainability,

signalling a shift towards more integrated and systemic design inquiry. Second, a thematic synthesis of 62 systematically selected core articles focused on CoPSS design was conducted. This synthesis aggregated the dominant research themes, employed methodological approaches, mobilised theoretical lenses, and proposed future research directions in the reviewed literature. Third, based on these findings, the study developed an integrative framework that maps the CoPSS design landscape across social, technical, and ecological dimensions, as well as their intersections. The findings indicate that although integration across these dimensions is beginning to emerge, research remains largely fragmented, with most studies addressing these aspects in isolation. By reconceptualising CoPSS design as a social–technical–ecological endeavour, the review underscores the need for more deliberate and systemic integration across design domains.

This paper makes the following contributions.

First, this review offers a critical reconceptualisation of CoPSS design as a multi-dimensional and interdependent domain, explicitly shaped by the co-evolution of servitisation, digitalisation, and sustainability transitions (Mourtzis et al., 2018a; Roman et al., 2023). While existing studies have typically treated these transitions as isolated influences or broad contextual drivers (Biedermann et al., 2022; Pirola et al., 2020), this review identifies them as mutually reinforcing structural shifts that shape CoPSS design logic at the system level. By using a social–technical–ecological lens, the review surfaces the fragmented treatment of these transitions and exposes underexplored intersections, such as the absence of lifecycle-oriented sustainability integration in digitally enabled service models or the disconnect between stakeholder governance and modular design strategies (Alcayaga and Hansen, 2025; Dober et al., 2023). This framing moves beyond thematic cataloguing by linking conceptual fragmentation to systemic sustainability challenges. In doing so, it strengthens the theoretical basis for understanding CoPSS design as a means of managing complexity across domains, and as a response mechanism to multiple interacting transitions rather than a reaction to isolated trends.

Second, the integrative framework developed in this review provides a theory-informed, multi-layered synthesis of the CoPSS design research landscape. It traces how dominant themes, literature-proposed directions, and gap-driven needs intersect across social, technical, and ecological dimensions. It identifies both thematic consolidations and areas of disjuncture, for example, the persistent separation between digital modelling and sustainability logics or between stakeholder-driven design and systemic coordination (Chang et al., 2023; Rondini et al., 2017). The framework contributes to field development by offering a coherent structure that facilitates shared vocabulary, encourages cross-domain collaboration, and supports mid-range theorising. It also clarifies how design priorities, such as stakeholder co-creation, real-time feedback, or modularisation (Naeem et al., 2025; Rashid et al., 2019), might be operationalised more holistically through lifecycle and governance linkages. As a result, the framework does not simply reflect existing literature but actively reorients it toward greater conceptual integration, promoting a more strategic, reflective, and systems-aware approach to CoPSS design research.

Third, this study makes a methodological contribution by demonstrating the value of combining keyword-based bibliometric mapping with interpretive thematic synthesis (Yin et al., 2020). While bibliometric methods offer macro-level insights into structural patterns and topic clustering, they risk overlooking conceptual nuance (Zupic and Cater, 2015). Conversely, thematic synthesis provides depth but can lack transparency or reproducibility (Torraco, 2016). This review demonstrates how a staged approach can overcome these limitations by aligning structural discovery with qualitative interpretation. Specifically, the use of bibliometric mapping informed article selection and interpretation, while the thematic synthesis enabled the grounded identification of research directions and methodological gaps. This dual approach addresses critiques that literature reviews in transdisciplinary fields such as CoPSS are often either too descriptive or too speculative.

By anchoring qualitative insight in structural evidence, the review enhances rigour and transparency. Moreover, the method is transferable, offering a robust template for future synthesis in other domains navigating complex multi-domain transitions.

In addition to these contributions, the integrative framework developed holds significant implications for both academic inquiry and design practice. Theoretically, it offers a structured lens for conceptualising CoPSS design as an interdependent process spanning social, technical, and ecological dimensions. This moves beyond conventional mono-dimensional approaches and supports interdisciplinary theorisation by revealing how stakeholder engagement, digital architectures, and sustainability objectives co-evolve within complex systems. It provides a platform for building mid-range theories that link design decisions with lifecycle dynamics and system-level transitions, which are currently under-theorised in the literature. From a practical standpoint, the framework functions as a diagnostic and developmental tool for practitioners seeking to identify integration blind spots and prioritise cross-domain collaboration. It facilitates structured reflection on where social, technological, and environmental considerations diverge or converge in design practice. This has direct utility for participatory design workshops, scenario planning, and lifecycle-based assessments. By embedding the logic of integration into design governance, the framework helps translate sustainability principles into actionable design strategies while fostering shared accountability across stakeholder groups.

Despite its contributions and implications, this paper has several limitations. First, the literature included in this study is limited to peer-reviewed academic journal articles indexed in Scopus. While this ensures scholarly rigour, it excludes valuable insights from industry reports, technical white papers, and practice-based case studies, which may reflect emerging or applied design knowledge not yet captured in academic discourse. Future studies should consider triangulating peer-reviewed sources with grey literature and practitioner insights to enhance contextual richness and practical relevance. Second, while this study provides a structured synthesis of design themes, methods, and theoretical framings in CoPSS design research, it does not empirically test or validate the proposed integrative framework. As such, its practical utility and scalability remain to be demonstrated. Further empirical research is needed to validate and adapt this framework through longitudinal case studies, participatory design experiments, or action research in real-world CoPSS projects. Finally, while this review proposes future research directions aligned with social–technical–ecological integration, it does not offer prescriptive tools or implementation strategies. The development of operational methods remains an open challenge. Future studies should move beyond identifying gaps to developing, testing, and refining actionable design instruments that address the structural and systemic demands of CoPSS design.

CRedit authorship contribution statement

Yiting Zhao: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Lihong Zhang:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Eunice Maytorena-Sanchez:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jun Zhang:** Writing – review & editing, Visualization, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yiting Zhao reports financial support was provided by China Scholarship Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the China Scholarship Council [grant number 202108440044].

Data availability

Data will be made available on request.

References

- Afshar, M., Wang, D., 2010. Systems thinking for designing sustainable product service systems: a case study using a system dynamics approach. *Des. Princ. Pract.* 4, 259–273.
- Alcayaga, A., Hansen, E.G., 2025. Smart circular economy as a service business model: an activity system framework and research agenda. *R. Manag.* 55, 508–530. <https://doi.org/10.1111/radm.12707>.
- Alcayaga, A., Wiener, M., Hansen, E.G., 2019. Towards a framework of smart-circular systems: An integrative literature review. *J. Clean. Prod.* 221, 622–634. <https://doi.org/10.1016/j.jclepro.2019.02.085>.
- Anke, J., 2019. Design-integrated financial assessment of smart services. *Electron. Mark.* 29, 19–35. <https://doi.org/10.1007/s12525-018-0300-y>.
- Annarelli, A., Battistella, C., Nonino, F., 2020. Competitive advantage implication of different product service system business models: consequences of 'not-replicable' capabilities. *J. Clean. Prod.* 247. <https://doi.org/10.1016/j.jclepro.2019.119121>.
- Belkadi, F., Boli, N., Usatorre, L., Maleki, E., Alexopoulos, K., Bernard, A., Mourtzis, D., 2020. A knowledge-based collaborative platform for PSS design and production. *CIRP J. Manuf. Sci. Technol.* 29, 220–231. <https://doi.org/10.1016/j.cirpj.2018.08.004>.
- Bertoni, A., Bertoni, M., 2020. PSS cost engineering: a model-based approach for concept design. *CIRP J. Manuf. Sci. Technol.* 29, 176–190. <https://doi.org/10.1016/j.cirpj.2018.08.001>.
- Bertoni, A., Bertoni, M., Panarotto, M., Johansson, C., Larsson, T.C., 2016. Value-driven product service systems development: methods and industrial applications. *CIRP J. Manuf. Sci. Technol.* 15, 42–55. <https://doi.org/10.1016/j.cirpj.2016.04.008>.
- Biedermann, A., López, N.M., Sáenz, J.L.S., Martín, J.I.V., 2022. Sustainability improvement in complex systems composed of products and services. *Int. J. Life Cycle Assess.* 27, 98–121. <https://doi.org/10.1007/s11367-021-02014-9>.
- Boess, S., 2022. Let's get sociotechnical: a design perspective on zero energy renovations. *Urban Plan* 7, 97–107. <https://doi.org/10.17645/up.v7i2.5107>.
- Bressanelli, G., Adrodegari, F., Perona, M., Sacconi, N., 2018. Exploring how usage-focused business models enable circular economy through digital technologies. *Sustain. Switz* 10. <https://doi.org/10.3390/su10030639>.
- Chang, Y., Ming, X., Chen, Z., Zhou, T., Liao, X., Song, W., 2023. Stakeholder requirement evaluation of smart industrial service ecosystem under pythagorean fuzzy environment for complex industrial contexts: a case study of renewable energy park. *Adv. Eng. Inform.* 55. <https://doi.org/10.1016/j.aei.2022.101823>.
- Chen, D., Chu, X., Yang, X., Sun, X., Li, Y., Su, Y., 2015. PSS solution evaluation considering sustainability under hybrid uncertain environments. *Expert Syst. Appl.* 42, 5822–5838. <https://doi.org/10.1016/j.eswa.2015.04.003>.
- Chuang, Y.-C., Chen, Y.M., 2022. Digital servitization of symbiotic service composition in product-service systems. *Comput. Ind.* 138. <https://doi.org/10.1016/j.compind.2022.103630>.
- Cong, J., Zheng, P., Bian, Y., Chen, C.-H., Li, J., Li, X., 2022. A machine learning-based iterative design approach to automate user satisfaction degree prediction in smart product-service system. *Comput. Ind. Eng.* 165. <https://doi.org/10.1016/j.cie.2022.107939>.
- Dehn, T., Chicksand, D., Knight, L., 2018. Verifying concepts for complex product-service system (COPSS) design. *Sustain. Technol. Innov. - Proc. 27th Annu. Conf. Int. Assoc. Manag. Technol. IAMOT* 2018.
- Dober, P., West, S., Wiesner, S.A., Ebel, M., 2023. Service lifecycle management in complex product-service systems. In: Alfnes, E., Romsdal, A., Strandhagen, J.O., von Cierninski, G., Romero, D. (Eds.), *IFIP Advances in Information and Communication Technology*. Presented at the IFIP Advances in Information and Communication Technology. Springer Science and Business Media, Deutschland GmbH, pp. 32–46. https://doi.org/10.1007/978-3-031-43666-6_3.
- Dong, L., Ren, M., Xiang, Z., Zheng, P., Cong, J., Chen, C.-H., 2023. A novel smart product-service system configuration method for mass personalization based on knowledge graph. *J. Clean. Prod.* 382. <https://doi.org/10.1016/j.jclepro.2022.135270>.
- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., Lim, W.M., 2021. How to conduct a bibliometric analysis: an overview and guidelines. *J. Bus. Res.* 133, 285–296. <https://doi.org/10.1016/j.jbusres.2021.04.070>.
- Eck, N. van, Waltman, L., 2009. Software survey: vosviewer, a computer program for bibliometric mapping. *Scientometrics* 84, 523–538. <https://doi.org/10.1007/s11192-009-0146-3>.
- Ferreira Junior, R., Scur, G., Nunes, B., 2022. Preparing for smart product-service system (PSS) implementation: an investigation into the daimler group. *Prod. Plann. Control* 33, 56–70. <https://doi.org/10.1080/09537287.2020.1821402>.
- García Lechuga, A., Cortés Robles, G., Arredondo Soto, K.C., Miranda Ackerman, M.A., 2023. The integration of the business model canvas and the service blueprinting to assist the conceptual design of new product-service systems. *J. Clean. Prod.* 415. <https://doi.org/10.1016/j.jclepro.2023.137801>.
- Geng, X., Chu, X., Xue, D., Zhang, Z., 2010. An integrated approach for rating engineering characteristics' final importance in product-service system development. *Comput. Ind. Eng.* 59, 585–594. <https://doi.org/10.1016/j.cie.2010.07.002>.
- Geng, X., Chu, X., Zhang, Z., 2012. An association rule mining and maintaining approach in dynamic database for aiding product-service system conceptual design. *Int. J. Adv. Manuf. Technol.* 62, 1–13. <https://doi.org/10.1007/s00170-011-3787-3>.
- Hallstedt, S.I., Isaksson, O., Rönnbäck, A.A.Ö., 2020. The need for new product development capabilities from digitalization, sustainability, and servitization trends. *Sustain. Switz* 12, 1–26. <https://doi.org/10.3390/su122310222>.
- Harrington, T.S., Srai, J.S., 2023. Designing a 'concept of operations' architecture for next-generation multi-organisational service networks. *AI Soc.* 38, 2533–2545. <https://doi.org/10.1007/s00146-016-0664-5>.
- Isaksson, O., Hallstedt, S.I., Rönnbäck, A.Ö., 2018. Digitalisation, sustainability and servitisation: consequences on product development capabilities in manufacturing firms. In: *Proc. Norddesign: Des. Era Digitalization, Norddesign*. Presented at the Proceedings of Norddesign: Design in the Era of Digitalization, Norddesign 2018, the Design Society.
- Kohtamäki, M., Rabetino, R., Parida, V., Sjödin, D., Henneberg, S., 2022. Managing digital servitization toward smart solutions: framing the connections between technologies, business models, and ecosystems. *Ind. Mark. Manag.* 105, 253–267. <https://doi.org/10.1016/j.indmarman.2022.06.010>.
- Li, A.Q., Kumar, M., Claes, B., Found, P., 2020. The state-of-the-art of the theory on product-service systems. *Int. J. Prod. Econ.* 222, 107491. <https://doi.org/10.1016/j.ijpe.2019.09.012>.
- Lugnet, J., Ericson, Å., Larsson, T., 2020. Design of product-service systems: toward an updated discourse. *Systems* 8, 1–14. <https://doi.org/10.3390/systems8040045>.
- Manzini, E., Vezzoli, C., 2003. A strategic design approach to develop sustainable product service systems: examples taken from the 'environmentally friendly innovation' Italian prize. *J. Clean. Prod., Product Service Systems and Sustainable Consumption* 11, 851–857. [https://doi.org/10.1016/S0959-6526\(02\)00153-1](https://doi.org/10.1016/S0959-6526(02)00153-1).
- May, G., Cho, S., Correia, A.T., Sifaka, R., Stokic, D., Kiritsis, D., 2022. Toward a reference terminology for product-service systems in the manufacturing domain. *Comput. Ind.* 142. <https://doi.org/10.1016/j.compind.2022.103729>.
- Medini, K., Boucher, X., 2019. Specifying a modelling language for PSS engineering – a development method and an operational tool. *Comput. Ind.* 108, 89–103. <https://doi.org/10.1016/j.compind.2019.02.014>.
- Meier, H., Roy, R., Seliger, G., 2010. Industrial product-service Systems—ips2. *CIRP Ann* 59, 607–627. <https://doi.org/10.1016/j.cirp.2010.05.004>.
- Mourtzis, D., Fotia, S., Boli, N., Pittaro, P., 2018a. Product-service system (PSS) complexity metrics within mass customization and industry 4.0 environment. *Int. J. Adv. Manuf. Technol.* 97, 91–103. <https://doi.org/10.1007/s00170-018-1903-3>.
- Mourtzis, D., Fotia, S., Boli, N., Vlachou, E., 2018b. An approach for the modelling and quantification of PSS customisation. *Int. J. Prod. Res.* 56, 1137–1153. <https://doi.org/10.1080/00207543.2017.1378956>.
- Naem, R., Kohtamäki, M., Parida, V., 2025. Artificial intelligence enabled product-service innovation: past achievements and future directions. *Rev. Manag. Sci.* 19, 2149–2192. <https://doi.org/10.1007/s11846-024-00757-x>.
- Pereira, V.R., Kreye, M.E., Carvalho, M.M., 2019. Customer-pulled and provider-pushed pathways for product-service system: the contingent effect of the business ecosystems. *J. Manuf. Technol. Manag.* 30, 729–747. <https://doi.org/10.1108/JMTM-07-2018-0209>.
- Pirola, F., Boucher, X., Wiesner, S., Pezzotta, G., 2020. Digital technologies in product-service systems: a literature review and a research agenda. *Comput. Ind.* 123. <https://doi.org/10.1016/j.compind.2020.103301>.
- Rahmati, P., Tafti, A., Westland, J.C., Hidalgo, C., 2021. When all products are digital: complexity and intangible value in the ecosystem of digitizing firms. *MIS Q. Manag. Inf. Syst.* 45, 1025–1058. <https://doi.org/10.25300/MISQ/2021/15384>.
- Rashid, Y., Waseem, A., Akbar, A.A., Azam, F., 2019. Value co-creation and social media: a systematic literature review using citation and thematic analysis. *Eur. Bus. Rev.* 31, 761–784. <https://doi.org/10.1108/EBR-05-2018-0106>.
- Ren, S., Shi, L., Liu, Y., Cai, W., Zhang, Y., 2023. A personalised operation and maintenance approach for complex products based on equipment portrait of product-service system. *Robot. Comput.-Integr. Manuf.* 80. <https://doi.org/10.1016/j.rcim.2022.102485>.
- Ring, G., Landsem, K.L., Holtskog, H., 2022. Design for excellence to explore complex product service systems: a case study. *Procedia CIRP*, 32nd CIRP Design Conference (CIRP Design 2022) - Design in a changing world 109, 690–694. <https://doi.org/10.1016/j.procir.2022.05.315>.
- Rittershaus, P., Renner, M., Aryan, V., 2023. A conceptual methodology to screen and adopt circular business models in small and medium scale enterprises (SMEs): a case study on child safety seats as a product service system. *J. Clean. Prod.* 390. <https://doi.org/10.1016/j.jclepro.2023.136083>.
- Rodríguez, A.E., Pezzotta, G., Pinto, R., Romero, D., 2020. A comprehensive description of the product-service systems' cost estimation process: an integrative review. *Int. J. Prod. Econ.* 221. <https://doi.org/10.1016/j.ijpe.2019.09.002>.
- Roehrich, J.K., Lewis, M.A., 2010. Towards a model of governance in complex (product-service) inter-organizational systems. *Construct. Manag. Econ.* 28, 1155–1164. <https://doi.org/10.1080/01446191003762249>.
- Roman, P., Thiry, G., Muylaert, C., Ruwet, C., Marchal, K., 2023. Defining and identifying strongly sustainable product-service systems (SSPS). *J. Clean. Prod.* 391, 136295. <https://doi.org/10.1016/j.jclepro.2023.136295>.
- Rondini, A., Tornese, F., Gnoni, M.G., Pezzotta, G., Pinto, R., 2017. Hybrid simulation modelling as a supporting tool for sustainable product service systems: a critical analysis. *Int. J. Prod. Res.* 55, 6932–6945. <https://doi.org/10.1080/00207543.2017.1330569>.

- Sadeghi, L., Dantan, J.-Y., Mathieu, L., Siadat, A., Aghelinejad, M.M., 2017. A design approach for safety based on product-service systems and function-behavior-structure. *CIRP J. Manuf. Sci. Technol.* 19, 44–56. <https://doi.org/10.1016/j.cirpj.2017.05.001>.
- Sakao, T., Hara, T., Fukushima, R., 2020. Using product/service-system family design for efficient customization with lean principles: model, method, and tool. *Sustain. Switz.* 12, 1–25. <https://doi.org/10.3390/su12145779>.
- Sakao, T., Song, W., Matschewsky, J., 2017. Creating service modules for customising product/service systems by extending DSM. *CIRP Ann. - Manuf. Technol.* 66, 21–24. <https://doi.org/10.1016/j.cirp.2017.04.107>.
- Sassanelli, C., Pezzotta, G., Pirola, F., Sala, R., Margarito, A., Lazoi, M., Corallo, A., Rossi, M., Terzi, S., 2018. Using design rules to guide the PSS design in an engineering platform based on the product service lifecycle management paradigm. *Int. J. Prod. Lifecycle Manag.* 11, 91–115. <https://doi.org/10.1504/IJPLM.2018.092826>.
- Sassanelli, C., Urbinati, A., Rosa, P., Chiaroni, D., Terzi, S., 2020. Addressing circular economy through design for X approaches: a systematic literature review. *Comput. Ind.* 120, 103245. <https://doi.org/10.1016/j.compind.2020.103245>.
- Scheepens, A.E., Vogtlander, J.G., Brezet, J.C., 2016. Two life cycle assessment (LCA) based methods to analyse and design complex (regional) circular economy systems. Case: making water tourism more sustainable. *J. Clean. Prod., Towards Post Fossil Carbon Societies: Regenerative and Preventative Eco-Industrial Development* 114, 257–268. <https://doi.org/10.1016/j.jclepro.2015.05.075>.
- Shimomura, Y., Nemoto, Y., Kimita, K., 2015. A method for analysing conceptual design process of product-service systems. *CIRP Ann. - Manuf. Technol.* 64, 145–148. <https://doi.org/10.1016/j.cirp.2015.04.035>.
- Snyder, H., 2019. Literature review as a research methodology: an overview and guidelines. *J. Bus. Res.* 104, 333–339. <https://doi.org/10.1016/j.jbusres.2019.07.039>.
- Son, H., Kwon, Y., Park, S.C., Lee, S., 2018. Using a design structure matrix to support technology roadmapping for product-service systems. *Technol. Anal. Strateg. Manag.* 30, 337–350. <https://doi.org/10.1080/09537325.2017.1310377>.
- Tong, Q., Ming, X., Zheng, M., Zhang, X., Wang, Z., 2022. An innovative model of smart product service ecosystem (SPSE) on sustainability: Survival system model, value emerges, and a case study. *Sustain. Switz.* 14. <https://doi.org/10.3390/su141912361>.
- Torraco, R.J., 2016. Writing integrative literature reviews: using the past and present to explore the future. *Hum. Resour. Dev. Rev.* 15, 404–428. <https://doi.org/10.1177/1534484316671606>.
- Tranfield, D., Denyer, D., Smart, P., 2003. Towards a methodology for developing evidence-informed management knowledge by means of systematic review. *Br. J. Manag.* 14, 207–222. <https://doi.org/10.1111/1467-8551.00375>.
- Tsunetomo, K., Watanabe, K., Kishita, Y., 2022. Smart product-service systems design process for socially conscious digitalization. *J. Clean. Prod.* 368. <https://doi.org/10.1016/j.jclepro.2022.133172>.
- Tukker, A., 2004. Eight types of product-service system: eight ways to sustainability? *Experiences from SusProNet. Bus. Strat. Environ.* 13, 246.
- Vezzoli, C., Ceschin, F., Diehl, J.C., Kohtala, C., 2015. New design challenges to widely implement 'Sustainable Product-Service Systems'. *J. Clean. Prod.* 97, 1–12. <https://doi.org/10.1016/j.jclepro.2015.02.061>. Special Volume: Why have 'Sustainable Product-Service Systems' not been widely implemented?
- Wang, X., Chen, X., Durugbo, C., Cai, Z., 2020. Manage risk of sustainable product-service systems: a case-based operations research approach. *Ann. Oper. Res.* 291, 897–920. <https://doi.org/10.1007/s10479-018-3051-4>.
- Wu, C., Chen, T., Li, Z., Liu, W., 2021. A function-oriented optimising approach for smart product service systems at the conceptual design stage: a perspective from the digital twin framework. *J. Clean. Prod.* 297. <https://doi.org/10.1016/j.jclepro.2021.126597>.
- Yang, X., Zhang, C., Li, Y., Tang, C., Liang, P., 2024. Dynamic user profile construction and its application to smart product-service system design: a maternity-oriented case study. *Expert Syst. Appl.* 249. <https://doi.org/10.1016/j.eswa.2024.123772>.
- Yin, D., Ming, X., Zhang, X., 2020. Sustainable and smart product innovation ecosystem: an integrative status review and future perspectives. *J. Clean. Prod.* 274, 123005. <https://doi.org/10.1016/j.jclepro.2020.123005>.
- Yip, M.H., Phaal, R., Probert, D.R., 2015. Characterising product-service systems in the healthcare industry. *Technol. Soc.* 43, 129–143. <https://doi.org/10.1016/j.techsoc.2015.05.014>.
- Zhang, K., Lin, K.-Y., Wang, J., Ma, Y., Li, H., Zhang, L., Liu, K., Feng, L., 2023. UNISON framework for user requirement elicitation and classification of smart product-service system. *Adv. Eng. Inform.* 57. <https://doi.org/10.1016/j.aei.2023.101996>.
- Zhang, P., Jing, S., Nie, Z., Zhao, B., Tan, R., 2021. Design and development of sustainable product service systems based on design-centric complexity. *Sustain. Switz.* 13, 1–27. <https://doi.org/10.3390/su13020532>.
- Zhang, Z., Xu, D., Ostrosi, E., Yu, L., Fan, B., 2019. A systematic decision-making method for evaluating design alternatives of product service system based on variable precision rough set. *J. Intell. Manuf.* 30, 1895–1909. <https://doi.org/10.1007/s10845-017-1359-6>.
- Zhao, Y., Xie, C., Lu, M., 2021. Design of smart building operations and maintenance management service system. In: Rebelo, F. (Ed.), *Advances in Ergonomics in Design, Lecture Notes in Networks and Systems*. Springer International Publishing, Cham, pp. 183–190. https://doi.org/10.1007/978-3-030-79760-7_23.
- Zheng, P., Chen, C.-H., Shang, S., 2019. Towards an automatic engineering change management in smart product-service systems – a DSM-Based learning approach. *Adv. Eng. Inform.* 39, 203–213. <https://doi.org/10.1016/j.aei.2019.01.002>.
- Zupic, I., Čater, T., 2015. Bibliometric methods in management and organization. *Organ. Res. Methods* 18, 429–472. <https://doi.org/10.1177/1094428114562629>.